|INCH-POUND|

MIL-M-28787/393
24 August 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

INTERFACE, BUS MODULE, KEY CODE AMJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital unidirectional bus interface module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the currentversion before use.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.55 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AMJ.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 100 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 + 5°C 4/		0 + 5°C 4/		60 + 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	S	1	INF	1	INF	1	INF	MOhm
B1-B16	Continuity	III	X	0 0	1.0	0.0	1.0	0 0	1.0	Ohm
B17-B32	Continuity	III	O	0.0	0.1	0 0	0.1	0 0	0.1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests--------						
D1-D16	Ibd	V	S	0	400	0	400	0	400	uA
E1-E64	Ios	VI	S	-225.0	-40.0	-225 0	-40.0	-225.0	-40.0	mA
F1-F16	Vic	VII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
G1-G64	Vol	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H64	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
H65,H66	Voh	IX	X	2.4	2.78	2.4	2.78	2.4	2.78	V
J1-J16	Iil	X	X	-800	0	-800	0	-800	0	uA
K1-K16	Iih	XI	X	0	80	0	80	0	80	uA
L1	Icc (low level)	XII	X	60.7	607.0	60.7	607.0	60 7	607.0	mA
L2	Icc (high level)	XII	X	39.1	391.0	39.1	391.0	39.1	391.0	mA
M1-M64	Propagation delay time (tPLH)(CL=150pF)	XIII	S	2	24	2	29	2	29	nS
M65-M128	Propagation delay time (tPLH)(CL=300pF)	XIII	O	2	33	2	38	2	38	nS

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
N1-N64	Propagation delay time (tPHL)(CL=150pF)	XIV	S	2	23	2	28	2	28	nS
N65-N128	Propagation delay time (tPHL)(CL=300pF)	XIV	Q	2	31	2	36	2	36	nS
P1-P64	Output crosstalk (low level)	XV	Q	0	400	0	400	0	400	mV
Q1-Q64	Output crosstalk (high level)	XVI	Q	0	400	0	400	0	400	mV
R	PSTA	XVII	Q	0	400	0	400	0	400	mV
S	PSFC	XVIII	S	3.85	7.15	3.85	7.15	3.85	7.15	uF
T1-T16	Cin	XIX	Q	2	55	2	55	2	55	pF

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:11Z
Check the source to verify that this is the current version before use.

1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/393-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling outputs is not permitted except to obtain extended drive capability. Each circuit whose output is paralleled with an output of another circuit must have each of its inputs paralleled with a corresponding input of the other circuits. Circuits may be paralleled only with other circuits within the same integrated circuit. The circuit groups are:

Group A:	Outputs D0/1A-D0/4A	Group I:	Outputs D0/1B-D0/4B
Group B:	Outputs D1/1A-D1/4A	Group J:	Outputs D1/1B-D1/4B
Group C:	Outputs D2/1A-D2/4A	Group K:	Outputs D2/1B-D2/4B
Group D:	Outputs D3/1A-D3/4A	Group L:	Outputs D3/1B-D3/4B
Group E:	Outputs D4/1A-D4/4A	Group M:	Outputs D4/1B-D4/4B
Group F:	Outputs D5/1A-D5/4A	Group N:	Outputs D5/1B-D5/4B
Group G:	Outputs D6/1A-D6/4A	Group O:	Outputs D6/1B-D6/4B
Group H:	Outputs D7/1A-D7/4A	Group P:	Outputs D7/1B-D7/4B

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-04-20T18:11Z
Check the source to verify that this is the current version before use.

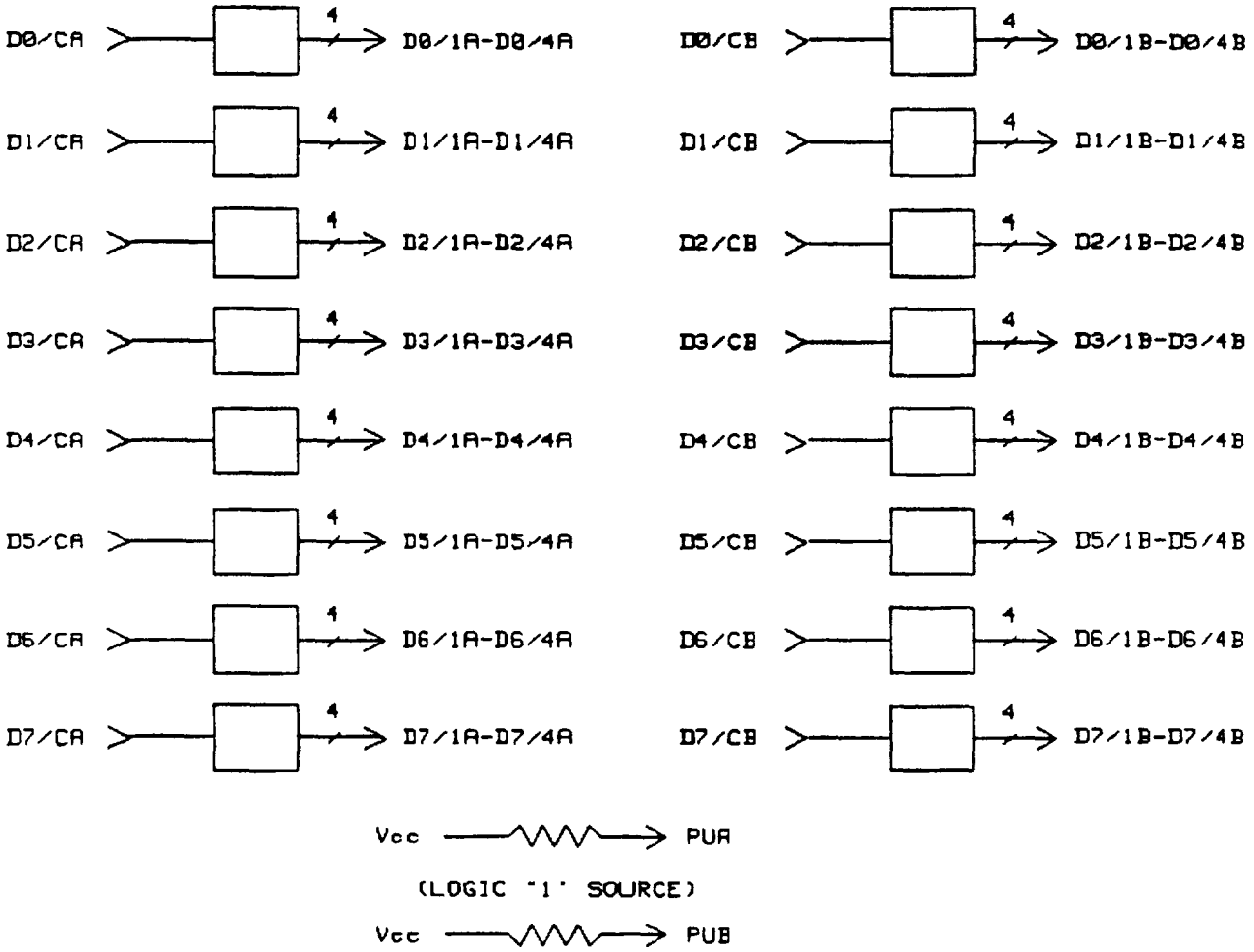


FIGURE 1. AMJ functional block diagram.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.

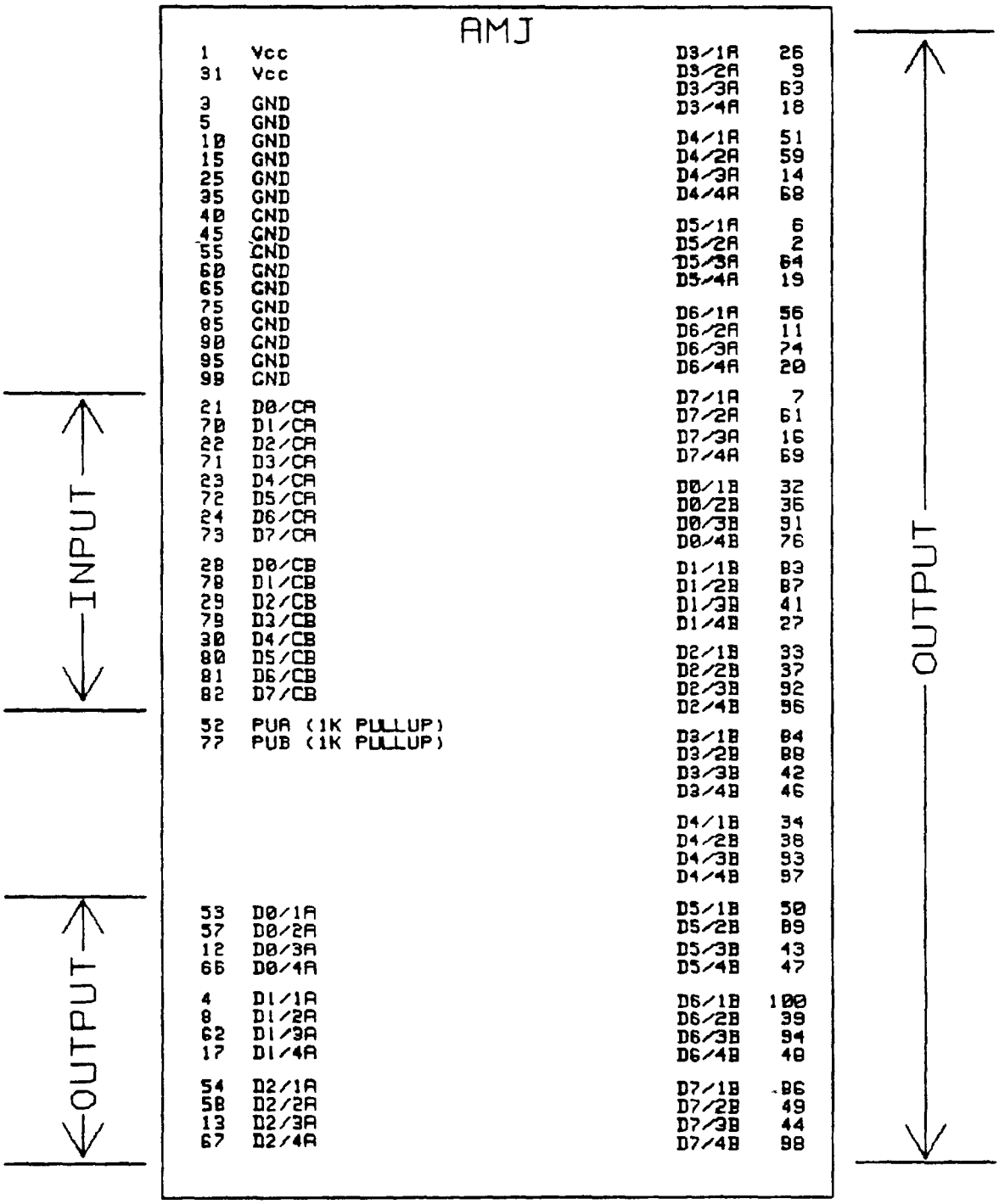


FIGURE 2. AMJ pin assignment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

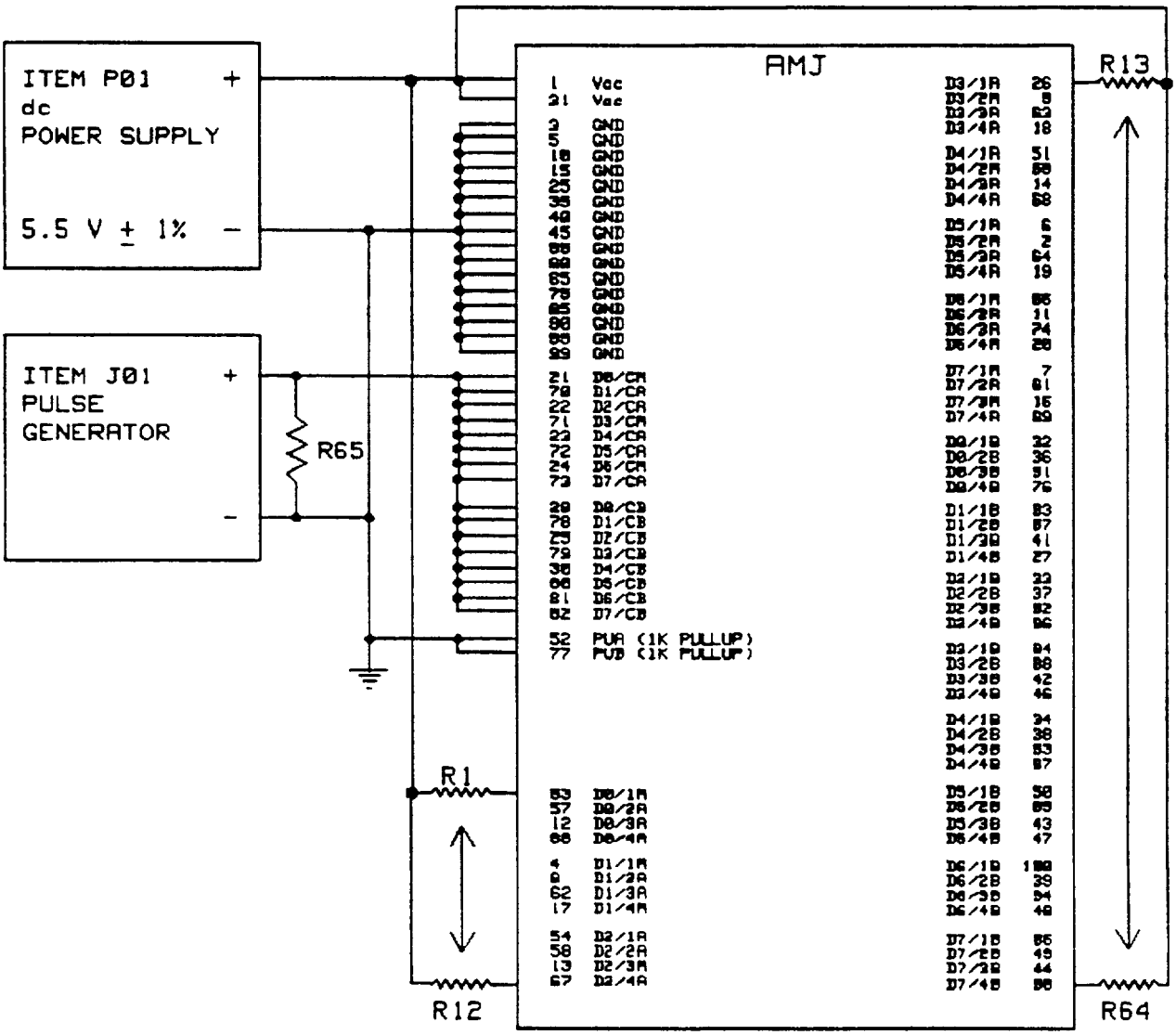
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, and G through S shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figure 3.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the currentversion before use.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition S, shall be performed using the initial limits at 25 ± 5°C ambient.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.



NOTES:

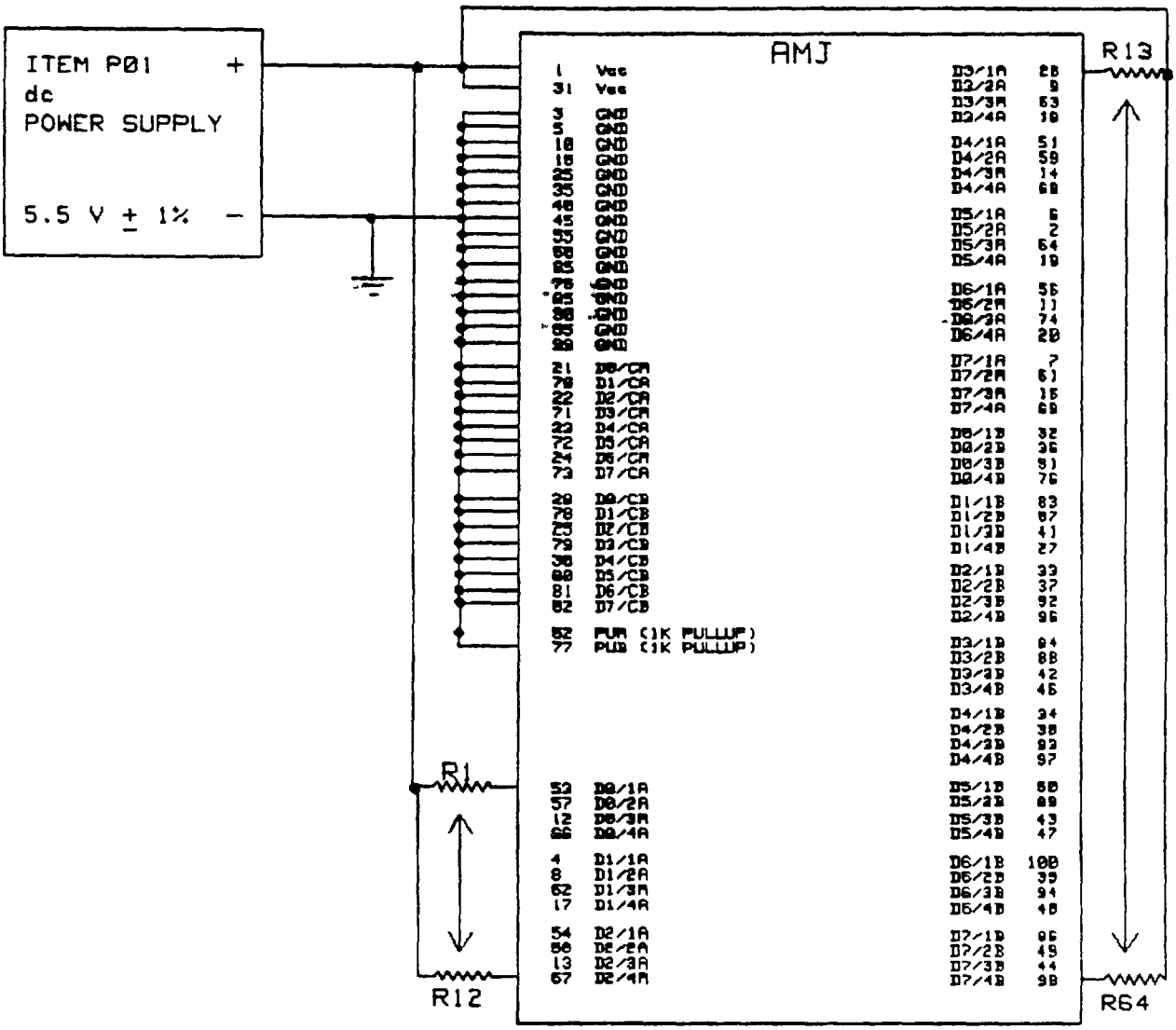
1. Item J01 setup: PA = 3.0 ± 0.1 V,
   PRP = 1.0 MHz ± 20 percent,
   PW = 500 ± 100 nS.

2. R1-R64 = 427 Ohm ± 1 percent, 1/8 W.

3. R65 = 51 Ohm ± 1 percent, 1 W.

FIGURE 3. AMJ operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.



NOTE:

1. R1-R64 = 427 Ohm ± 1 percent, 1/8 W.

FIGURE 4. AMJ component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

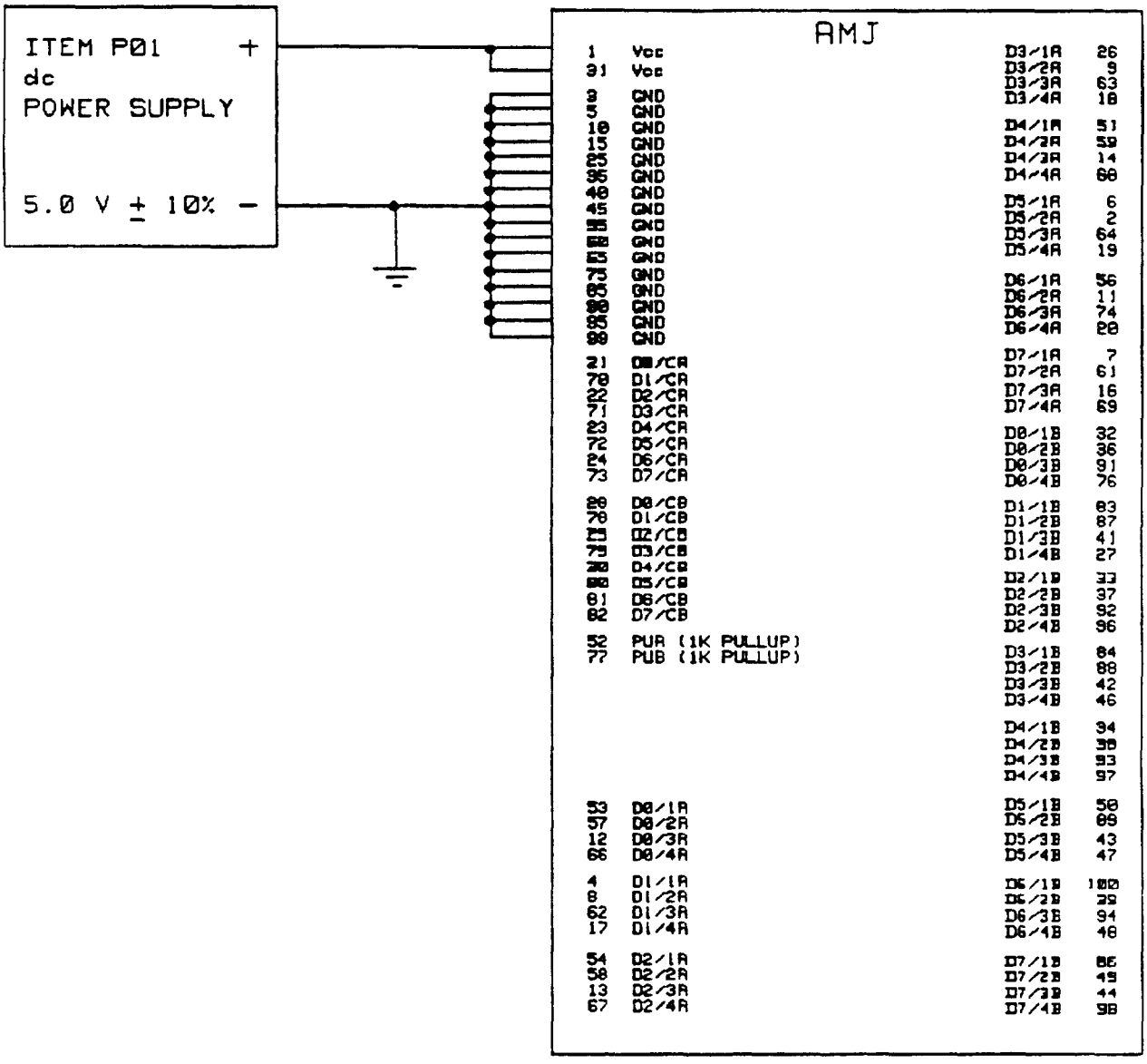


FIGURE 5. AMJ vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:11Z
Check the source to verify that this is the current version before use.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B and G through S shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B and G through S shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.338 W (max); 1.806 W (typical).

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D0/CA	-1.5	0.7	2.0	-800	80	400	55
D1/CA	-1.5	0.7	2.0	-800	80	400	55
D2/CA	-1.5	0.7	2.0	-800	80	400	55
D3/CA	-1.5	0.7	2.0	-800	80	400	55
D4/CA	-1.5	0.7	2.0	-800	80	400	55
D5/CA	-1.5	0.7	2.0	-800	80	400	55
D6/CA	-1.5	0.7	2.0	-800	80	400	55
D7/CA	-1.5	0.7	2.0	-800	80	400	55
D0/CB	-1.5	0.7	2.0	-800	80	400	55
D1/CB	-1.5	0.7	2.0	-800	80	400	55
D2/CB	-1.5	0.7	2.0	-800	80	400	55
D3/CB	-1.5	0.7	2.0	-800	80	400	55
D4/CB	-1.5	0.7	2.0	-800	80	400	55
D5/CB	-1.5	0.7	2.0	-800	80	400	55

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D6/CB	-1.5	0.7	2.0	-800	80	400	55
D7/CB	-1.5	0.7	2.0	-800	80	400	55

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios min (mA)	Cout max (pF)
D0/1A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D0/2A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D0/3A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D0/4A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D1/1A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D1/2A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D1/3A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D1/4A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D2/1A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D2/2A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios min (mA)	Cout max (pF)
D2/3A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D2/4A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D3/1A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D3/2A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D3/3A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D3/4A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D4/1A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D4/2A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D4/3A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D4/4A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D5/1A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D5/2A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D5/3A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D5/4A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios min (mA)	Cout max (pF)
D6/1A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D6/2A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D6/3A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D6/4A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D7/1A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D7/2A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D7/3A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D7/4A	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D0/1B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D0/2B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D0/3B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D0/4B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D1/1B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D1/2B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios min (mA)	Cout max (pF)
D1/3B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D1/4B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D2/1B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D2/2B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D2/3B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D2/4B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D3/1B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D3/2B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D3/3B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D3/4B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D4/1B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D4/2B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D4/3B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D4/4B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios min (mA)	Cout max (pF)
D5/1B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D5/2B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D5/3B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D5/4B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D6/1B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D6/2B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D6/3B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D6/4B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D7/1B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D7/2B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D7/3B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA
D7/4B	2.4	0.4	-3.0	12.0	NA	NA	-225	NA

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module contains two 8-bit noninverting, buffer sections to provide expanded drive capacity, bus separation, and isolation between other computer components.

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II, figure 6.

(b) Continuity test - table III, figure 7.

(c) Vcc stress test - table IV, figure 8.

(d) Ibd test - table V, figure 9.

(e) Ios test - table VI, figure 10.

(f) Vic test - table VII, figure 11.

(g) Vol test - table VIII, figures 12 and 13.

(h) Voh test - table IX, figures 12 and 13.

(i) Iil test - table X, figure 14.

(j) Iih test - table XI, figure 14.

(k) Icc test - table XII, figure 15.

(l) Propagation delay time (tPLH) test - table XIII, figures 16 and 17.

(m) Propagation delay time (tPHL) test - table XIV, figures 16 and 17.

(n) Output crosstalk, low level, test - table XV, figures 18 and 19.

(o) Output crosstalk, high level, test - table XVI, figures 18 and 19.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

(p) PSTA test - table XVII, figures 20 and 21.

(q) PSFC test - table XVIII, figure 22.

(r) Cin test - table XIX, figure 22.

6.6 Functional test vectors. Functional test vectors are inherent in the test tables.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0065)

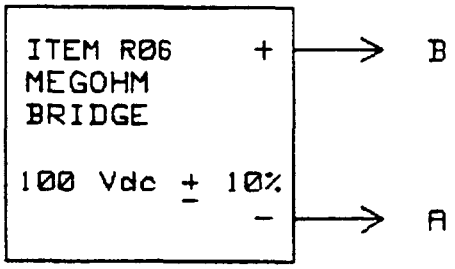
Agent: NW

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
A	Isolation	All pins	Frame	10	INF	MOhm	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.



AMJ

1	Vcc	D3/1A	26
31	Vcc	D3/2A	9
		D3/3A	53
3	GND	D3/4A	18
5	GND		
10	GND	D4/1A	51
15	GND	D4/2A	89
25	GND	D4/3A	14
35	GND	D4/4A	60
40	GND		
45	GND	D5/1A	5
55	GND	D5/2A	2
60	GND	D5/3A	64
65	GND	D5/4A	15
75	GND		
85	GND	D6/1A	56
90	GND	D6/2A	11
95	GND	D6/3A	74
98	GND	D6/4A	20
21	D0/CA	D7/1A	7
70	D1/CA	D7/2A	61
22	D2/CA	D7/3A	16
71	D3/CA	D7/4A	69
23	D4/CA	D0/1B	32
72	D5/CA	D0/2B	36
24	D6/CA	D0/3B	91
73	D7/CA	D0/4B	76
28	D0/CB	D1/1B	83
78	D1/CB	D1/2B	87
29	D2/CB	D1/3B	41
79	D3/CB	D1/4B	27
30	D4/CB		
80	D5/CB	D2/1B	33
81	D6/CB	D2/2B	37
82	D7/CB	D2/3B	92
		D2/4B	96
52	PUA (1K PULLUP)		
77	PUB (1K PULLUP)	D3/1B	84
		D3/2B	88
		D3/3B	42
		D3/4B	46
		D4/1B	34
		D4/2B	38
		D4/3B	93
		D4/4B	97
53	D0/1A	D5/1B	50
57	D0/2A	D5/2B	89
12	D0/3A	D5/3B	43
86	D0/4A	D5/4B	47
4	D1/1A	D6/1B	100
8	D1/2A	D6/2B	39
62	D1/3A	D6/3B	94
17	D1/4A	D6/4B	48
54	D2/1A	D7/1B	86
58	D2/2A	D7/2B	45
13	D2/3A	D7/3B	44
67	D2/4A	D7/4B	98

NOTES:

1. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

2. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

FIGURE 6. Isolation test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE III. Continuity test.

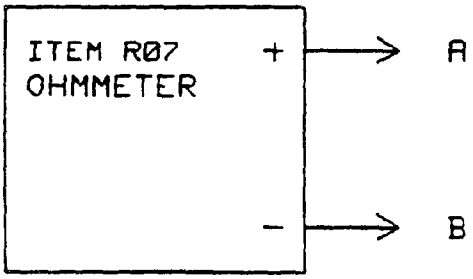
Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	7
B2	Continuity, GND	10	15				
B3	Continuity, GND	10	25				
B4	Continuity, GND	10	40				
B5	Continuity, GND	10	3				
B6	Continuity, GND	10	5				
B7	Continuity, GND	10	35				
B8	Continuity, GND	10	45				
B9	Continuity, GND	10	55				
B10	Continuity, GND	10	60				
B11	Continuity, GND	10	65				
B12	Continuity, GND	10	75				
B13	Continuity, GND	10	85				
B14	Continuity, GND	10	90				
B15	Continuity, GND	10	95				
B16	Continuity, GND	10	99	0.0	1.0	Ohm	7
B17	Continuity, Vcc	1	31	0.0	0.1	Ohm	7
B18	Continuity, GND	10	15				
B19	Continuity, GND	10	25				
B20	Continuity, GND	10	40	0.0	0.1	Ohm	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE III. Continuity test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
B21	Continuity, GND	10	3	0.0	0.1	Ohm	7
B22	Continuity, GND	10	5				
B23	Continuity, GND	10	35				
B24	Continuity, GND	10	45				
B25	Continuity, GND	10	55				
B26	Continuity, GND	10	60				
B27	Continuity, GND	10	65				
B28	Continuity, GND	10	75				
B29	Continuity, GND	10	85				
B30	Continuity, GND	10	90				
B31	Continuity, GND	10	95				
B32	Continuity, GND	10	99	0.0	0.1	Ohm	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.



AMJ

1	Vcc	D3/1A	26
81	Vcc	D3/2A	9
		D3/3A	63
3	GND	D3/4A	18
5	GND		
10	GND	D4/1A	51
15	GND	D4/2A	59
25	GND	D4/3A	14
35	GND	D4/4A	68
40	GND		
45	GND	D5/1A	6
55	GND	D5/2A	2
60	GND	D5/3A	64
65	GND	D5/4A	19
75	GND		
83	GND	D6/1A	56
88	GND	D6/2A	11
95	GND	D6/3A	74
99	GND	D6/4A	20
		D7/1A	7
21	D0/CA	D7/2A	61
70	D1/CA	D7/3A	16
22	D2/CA	D7/4A	69
71	D3/CA		
23	D4/CA	D0/1B	32
72	D5/CA	D0/2B	36
24	D6/CA	D0/3B	91
73	D7/CA	D0/4B	76
20	D0/CB	D1/1B	83
78	D1/CB	D1/2B	87
23	D2/CB	D1/3B	41
79	D3/CB	D1/4B	27
80	D4/CB		
80	D5/CB	D2/1B	33
81	D6/CB	D2/2B	37
82	D7/CB	D2/3B	92
		D2/4B	96
52	PUR (1K PULLUP)		
77	PUB (1K PULLUP)	D3/1B	84
		D3/2B	88
		D3/3B	42
		D3/4B	46
		D4/1B	34
		D4/2B	38
		D4/3B	93
		D4/4B	97
53	D0/1A	D5/1B	50
57	D0/2A	D5/2B	89
12	D0/3A	D5/3B	43
66	D0/4A	D5/4B	47
4	D1/1A	D6/1B	100
8	D1/2A	D6/2B	39
62	D1/3A	D6/3B	94
17	D1/4A	D6/4B	48
54	D2/1A	D7/1B	86
58	D2/2A	D7/2B	45
13	D2/3A	D7/3B	44
67	D2/4A	D7/4B	98

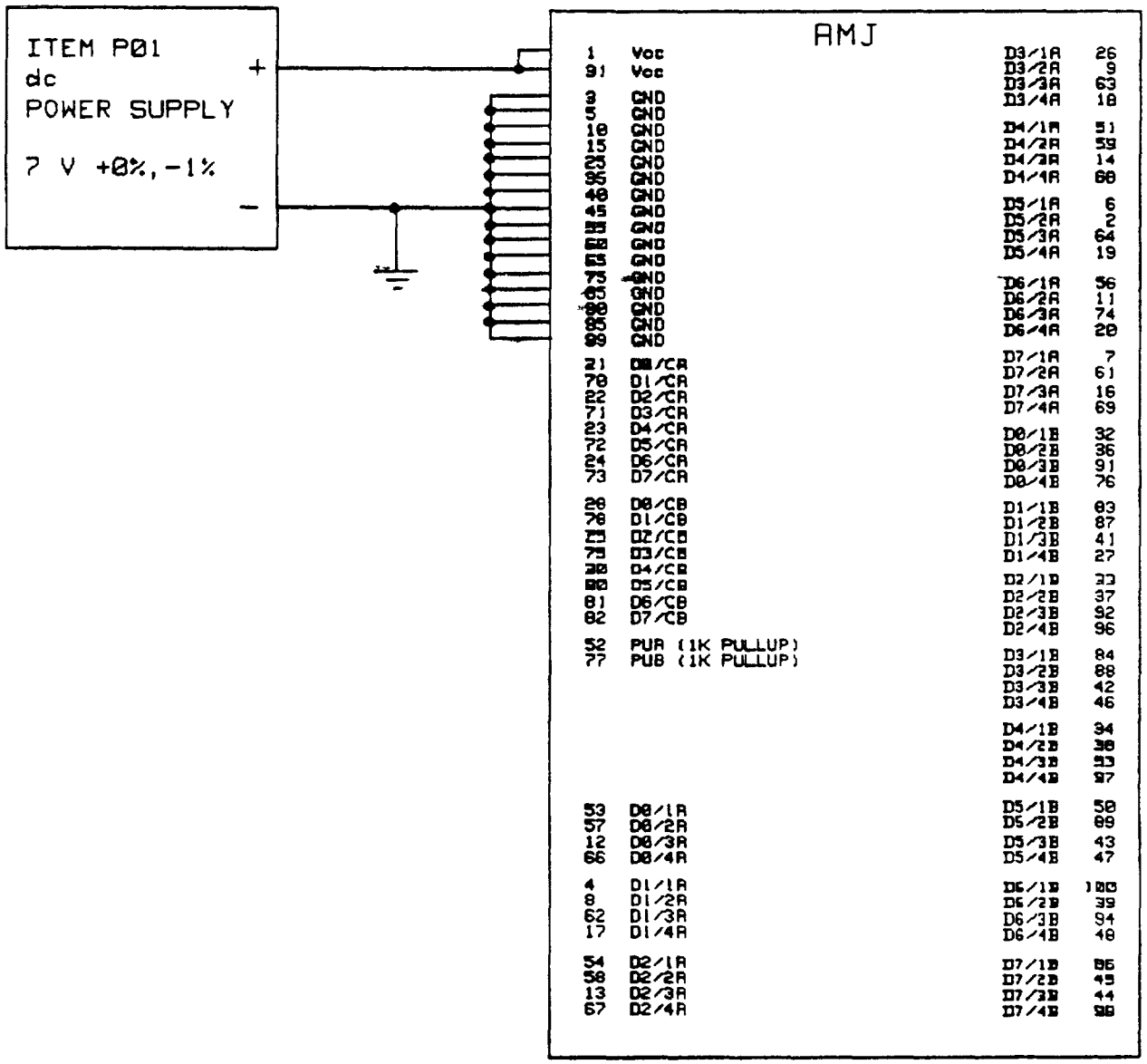
FIGURE 7. Continuity test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE IV. Power supply stress test.

Test cond	Power supply	25 ± 5°C Initial limits	Fig
C	Vcc Stress	Module shall pass all remaining tests.	8

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.



NOTE:

1. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

FIGURE 8. Vcc stress test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

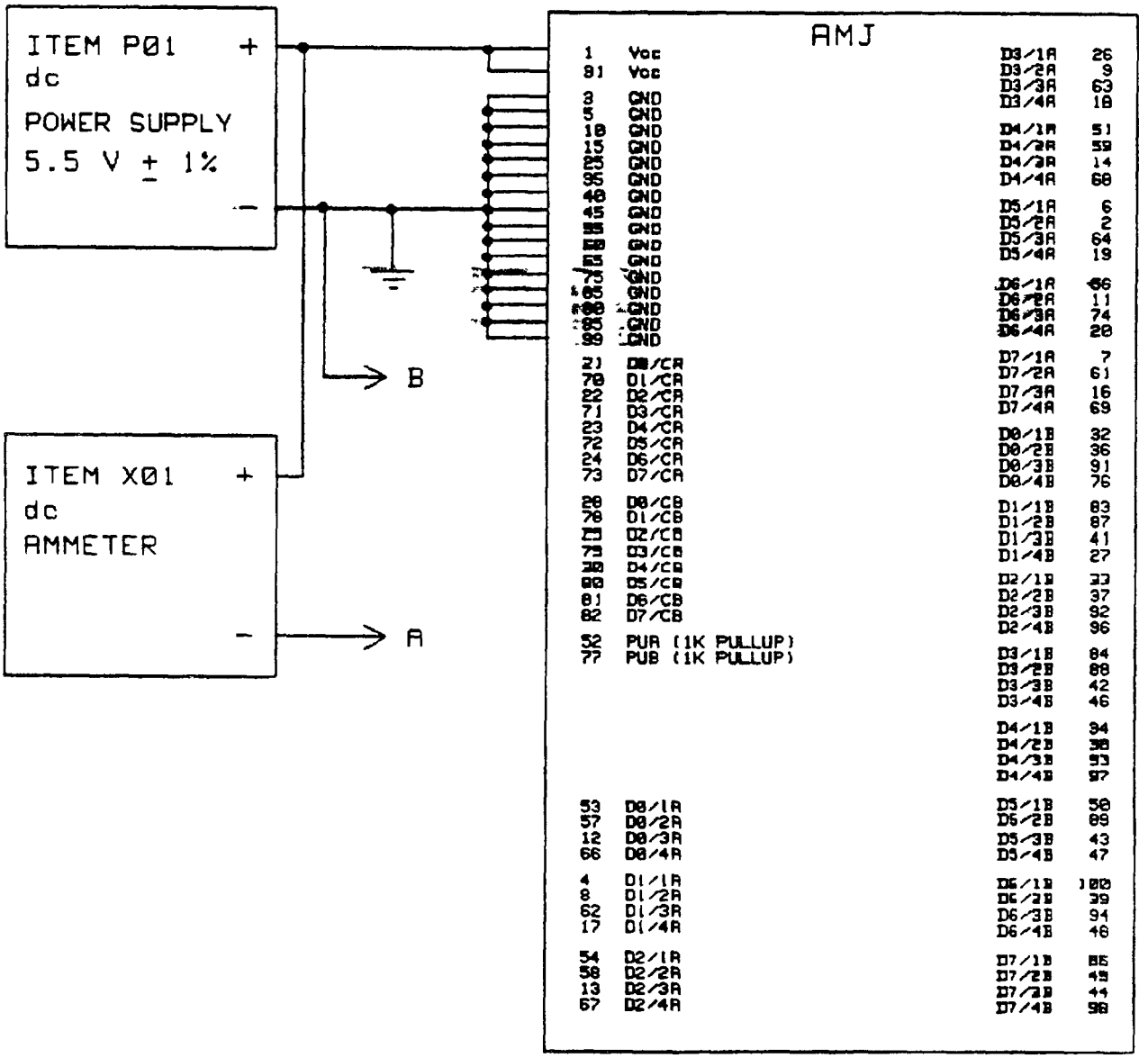
TABLE V. Input breakdown current test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
D1	D0/CA	21	See note 1	0	400	uA	9
D2	D1/CA	70					
D3	D2/CA	22					
D4	D3/CA	71					
D5	D4/CA	23					
D6	D5/CA	72					
D7	D6/CA	24					
D8	D7/CA	73					
D9	D0/CB	28					
D10	D1/CB	78					
D11	D2/CB	29					
D12	D3/CB	79					
D13	D4/CB	30					
D14	D5/CB	80					
D15	D6/CB	81					
D16	D7/CB	82	See note 1	0	400	uA	9

NOTE:

1. One input pin will be connected to point A, while all of the remaining input pins shall be connected to point B. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81 and 82.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.



NOTE:

1. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 9. Input breakdown current test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the currentversion before use.

TABLE VI. Output short circuit current test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
E1	D0/1A	53	See note 1	21	-225.0	-40.0	mA	10
E2	D0/2A	57						
E3	D0/3A	12						
E4	D0/4A	66		21				
E5	D1/1A	4		70				
E6	D1/2A	8						
E7	D1/3A	62						
E8	D1/4A	17		70				
E9	D2/1A	54		22				
E10	D2/2A	58						
E11	D2/3A	13						
E12	D2/4A	67		22				
E13	D3/1A	26		71				
E14	D3/2A	9						
E15	D3/3A	63						
E16	D3/4A	18		71				
E17	D4/1A	51		23				
E18	D4/2A	59						
E19	D4/3A	14						
E20	D4/4A	68	See note 1	23	-225.0	-40.0	mA	10

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
E21	D5/1A	6	See note 1	72	-225.0	-40.0	mA	10
E22	D5/2A	2						
E23	D5/3A	64						
E24	D5/4A	19		72				
E25	D6/1A	56		24				
E26	D6/2A	11						
E27	D6/3A	74						
E28	D6/4A	20		24				
E29	D7/1A	7		73				
E30	D7/2A	61						
E31	D7/3A	16						
E32	D7/4A	69		73				
E33	D0/1B	32		28				
E34	D0/2B	36						
E35	D0/3B	91						
E36	D0/4B	76		28				
E37	D1/1B	83		78				
E38	D1/2B	87						
E39	D1/3B	41						
E40	D1/4B	27	See note 1	78	-225.0	-40.0	mA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
E41	D2/1B	33	See note 1	29	-225.0	-40.0	mA	10
E42	D2/2B	37						
E43	D2/3B	92						
E44	D2/4B	96		29				
E45	D3/1B	84		79				
E46	D3/2B	88						
E47	D3/3B	42						
E48	D3/4B	46		79				
E49	D4/1B	34		30				
E50	D4/2B	38						
E51	D4/3B	93						
E52	D4/4B	97		30				
E53	D5/1B	50		80				
E54	D5/2B	89						
E55	D5/3B	43						
E56	D5/4B	47	See note 1	80	-225.0	-40.0	mA	10
E57	D6/1B	100	See note 1	81	-225.0	-40.0	mA	10
E58	D6/2B	39						
E59	D6/3B	94						
E60	D6/4B	48	See note 1	81	-225.0	-40.0	mA	10

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

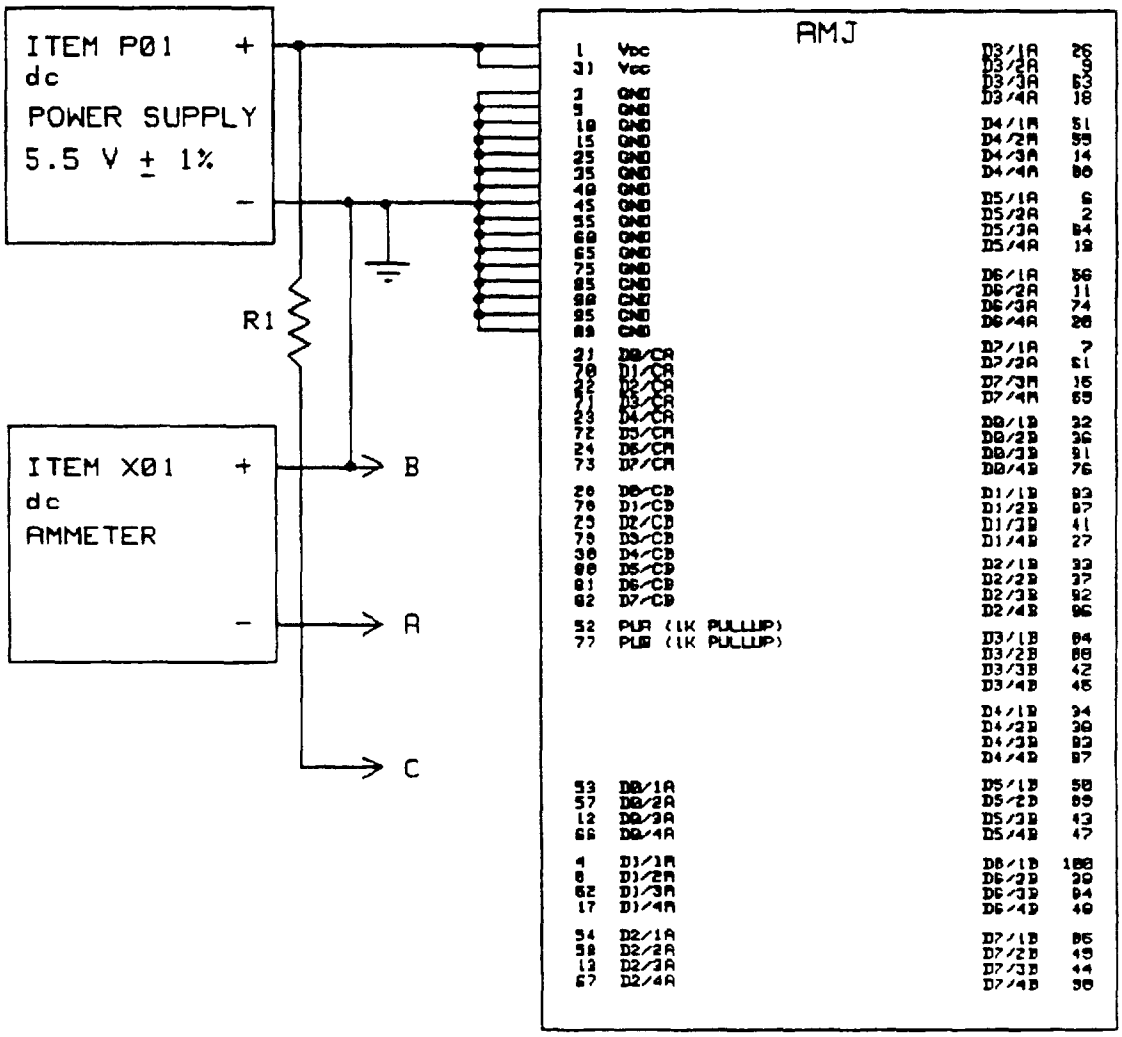
TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
E61	D7/1B	86	See note 1	82	-225.0	-40.0	mA	10
E62	D7/2B	49						
E63	D7/3B	44						
E64	D7/4B	98	See note 1	82	-225.0	-40.0	mA	10

NOTE:

1. One input pin will be connected to point C, while all of the remaining input pins shall be connected to point B. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81 and 82.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.



NOTES:

1. Do not isolate the chassis of X01 from earth ground.

2. When performing the output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second.

3. R1 = 1 KOhm ± 5 percent, 1/4 W.

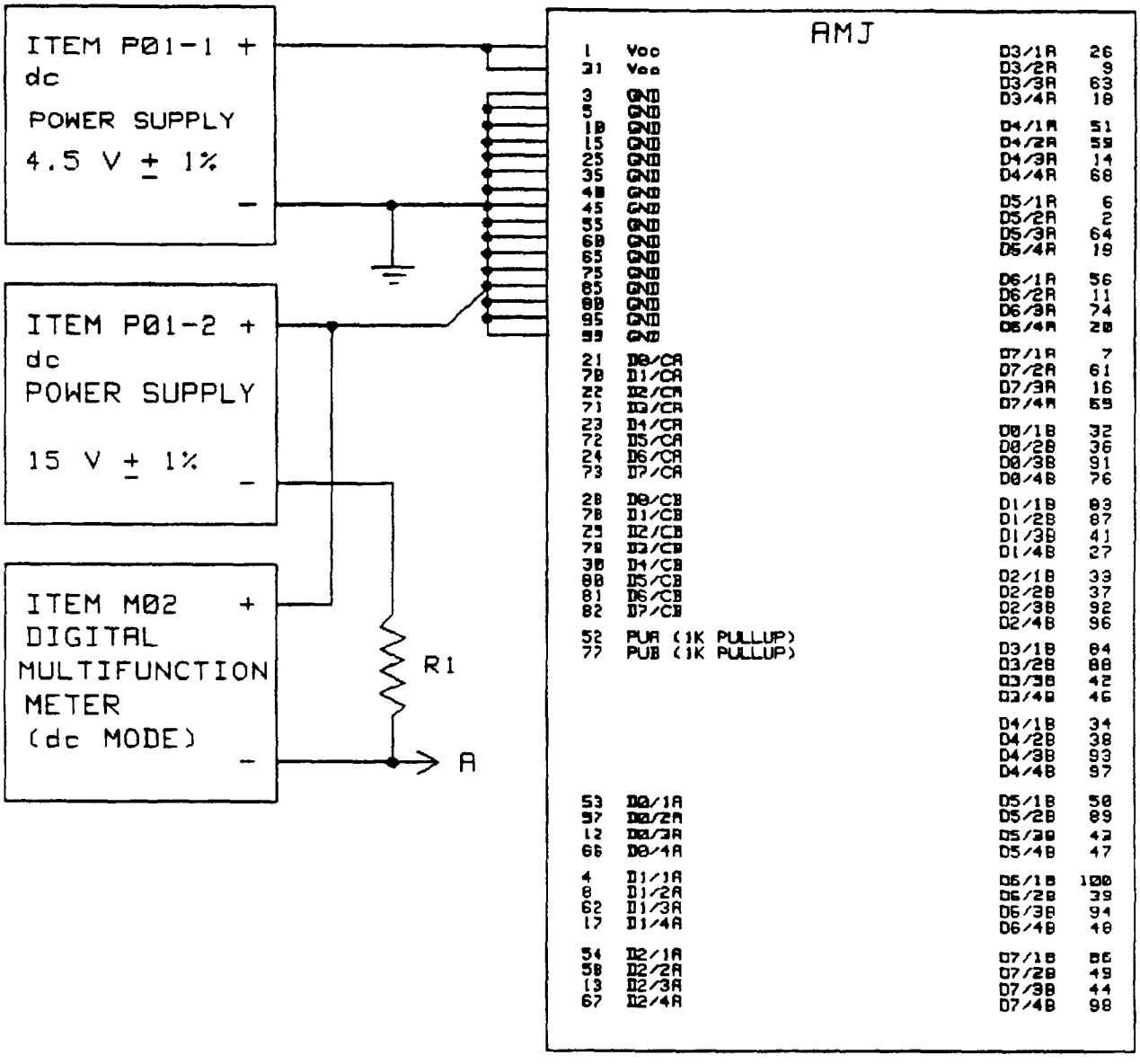
FIGURE 10. Output short circuit current test for table VI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE VII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
F1	D0/CA	21	-1.5	0.0	V	11
F2	D1/CA	70				
F3	D2/CA	22				
F4	D3/CA	71				
F5	D4/CA	23				
F6	D5/CA	72				
F7	D6/CA	24				
F8	D7/CA	73				
F9	D0/CB	28				
F10	D1/CB	78				
F11	D2/CB	29				
F12	D3/CB	79				
F13	D4/CB	30				
F14	D5/CB	80				
F15	D6/CB	81				
F16	D7/CB	82	-1.5	0.0	V	11

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.



NOTE:

1. R1 = 750 Ohm ± 1 percent, 1/4 W.

FIGURE 11. Input clamp diode voltage test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
G1	D0/1A	53	Open	See note 1	0.0	0.4	V	12
G2	D0/2A	57						
G3	D0/3A	12						
G4	D0/4A	66						
G5	D1/1A	4						
G6	D1/2A	8						
G7	D1/3A	62						
G8	D1/4A	17						
G9	D2/1A	54						
G10	D2/2A	58						
G11	D2/3A	13						
G12	D2/4A	67						
G13	D3/1A	26						
G14	D3/2A	9						
G15	D3/3A	63						
G16	D3/4A	18						
G17	D4/1A	51						
G18	D4/2A	59						
G19	D4/3A	14						
G20	D4/4A	68						
G21	D5/1A	6	Open	See note 1	0.0	0.4	V	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
G22	D5/2A	2	Open	See note 1	0.0	0.4	V	12
G23	D5/3A	64						
G24	D5/4A	19						
G25	D6/1A	56						
G26	D6/2A	11						
G27	D6/3A	74						
G28	D6/4A	20						
G29	D7/1A	7						
G30	D7/2A	61						
G31	D7/3A	16						
G32	D7/4A	69						
G33	D0/1B	32						
G34	D0/2B	36						
G35	D0/3B	91						
G36	D0/4B	76						
G37	D1/1B	83						
G38	D1/2B	87						
G39	D1/3B	41						
G40	D1/4B	27						
G41	D2/1B	33	Open	See note 1	0.0	0.4	V	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
G42	D2/2B	37	Open	See note 1	0.0	0.4	V	12
G43	D2/3B	92						
G44	D2/4B	96						
G45	D3/1B	84						
G46	D3/2B	88						
G47	D3/3B	42						
G48	D3/4B	46						
G49	D4/1B	34						
G50	D4/2B	38						
G51	D4/3B	93						
G52	D4/4B	97						
G53	D5/1B	50						
G54	D5/2B	89						
G55	D5/3B	43						
G56	D5/4B	47						
G57	D6/1B	100						
G58	D6/2B	39						
G59	D6/3B	94						
G60	D6/4B	48						
G61	D7/1B	86						
G62	D7/2B	49	Open	See note 1	0.0	0.4	V	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
G63	D7/3B	44	Open	See note 1	0.0	0.4	V	12
G64	D7/4B	98	Open	See note 1	0.0	0.4	V	12

NOTE:

1. All of the remaining input pins shall be connected to point C. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81 and 82.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
H1	D0/1A	53	21	See note 1	2.4	4.5	V	12
H2	D0/2A	57						
H3	D0/3A	12						
H4	D0/4A	66	21					
H5	D1/1A	4	70					
H6	D1/2A	8						
H7	D1/3A	62						
H8	D1/4A	17	70					
H9	D2/1A	54	22					
H10	D2/2A	58						
H11	D2/3A	13						
H12	D2/4A	67	22					
H13	D3/1A	26	71					
H14	D3/2A	9						
H15	D3/3A	63						
H16	D3/4A	18	71					
H17	D4/1A	51	23					
H18	D4/2A	59						
H19	D4/3A	14						
H20	D4/4A	68	23	See note 1	2.4	4.5	V	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
H21	D5/1A	6	72	See note 1	2.4	4.5	V	12
H22	D5/2A	2						
H23	D5/3A	64						
H24	D5/4A	19	72					
H25	D6/1A	56	24					
H26	D6/2A	11						
H27	D6/3A	74						
H28	D6/4A	20	24					
H29	D7/1A	7	73					
H30	D7/2A	61						
H31	D7/3A	16						
H32	D7/4A	69	73					
H33	D0/1B	32	28					
H34	D0/2B	36						
H35	D0/3B	91						
H36	D0/4B	76	28					
H37	D1/1B	83	78					
H38	D1/2B	87						
H39	D1/3B	41						
H40	D1/4B	27	78	See note 1	2.4	4.5	V	12

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
H41	D2/1B	33	29	See note 1	2.4	4.5	V	12
H42	D2/2B	37						
H43	D2/3B	92						
H44	D2/4B	96	29					
H45	D3/1B	84	79					
H46	D3/2B	88						
H47	D3/3B	42						
H48	D3/4B	46	79					
H49	D4/1B	34	30					
H50	D4/2B	38						
H51	D4/3B	93						
H52	D4/4B	97	30					
H53	D5/1B	50	80					
H54	D5/2B	89						
H55	D5/3B	43						
H56	D5/4B	47	80					
H57	D6/1B	100	81					
H58	D6/2B	39						
H59	D6/3B	94						
H60	D6/4B	48	81	See note 1	2.4	4.5	V	12

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:11Z
Check the source to verify that this is the current version before use.

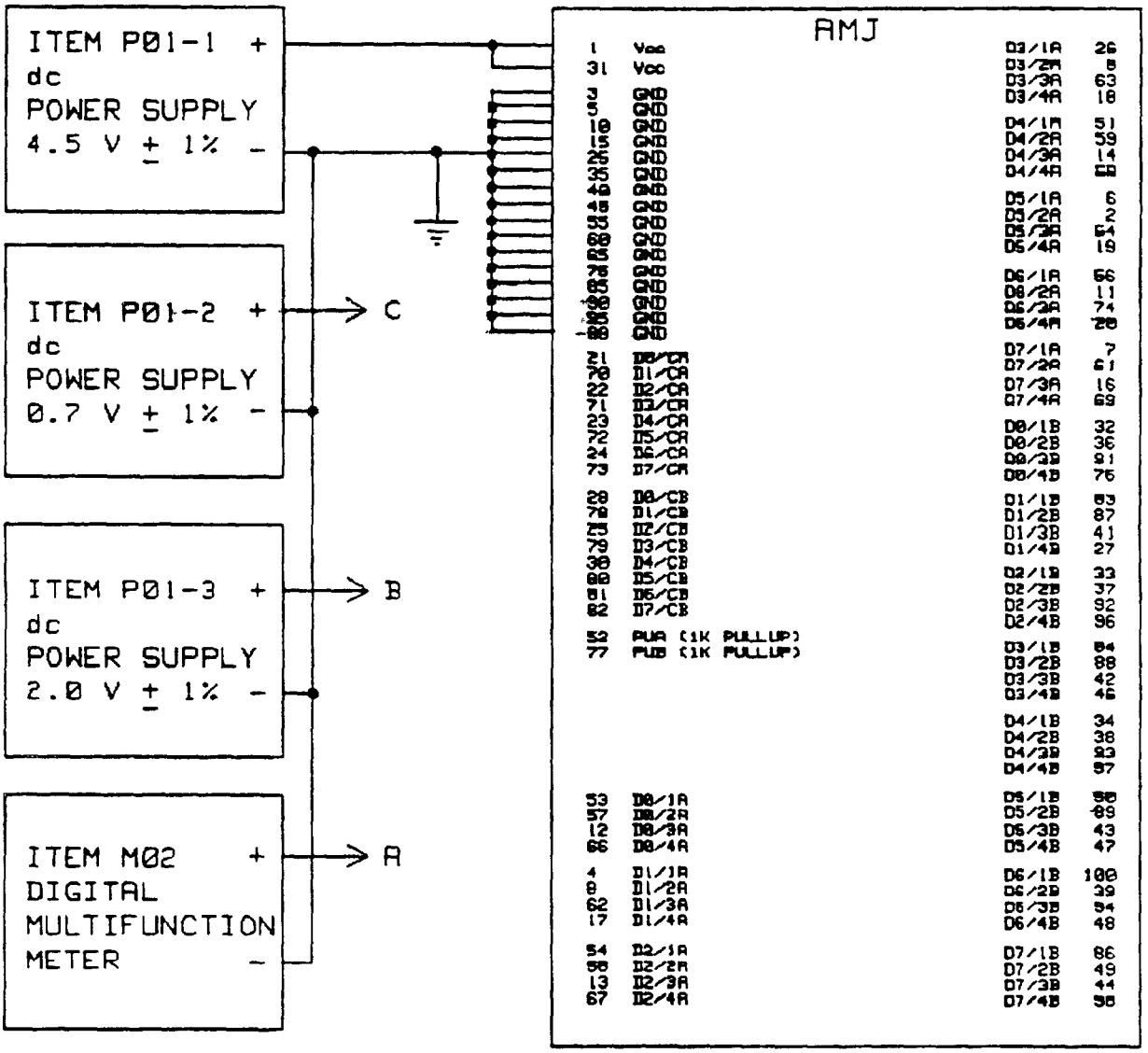
TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
H61	D7/1B	86	82	See note 1	2.4	4.5	V	12
H62	D7/2B	49						
H63	D7/3B	44						
H64	D7/4B	98	82		2.4	4.5	V	12
H65	PUA	52	Open		2.4	2.78	V	12
H66	PUB	77	Open	See note 1	2.4	2.78	V	12

NOTE:

1. All of the remaining input pins shall be connected to point C. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81 and 82.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:11Z
Check the source to verify that this is the current version before use.

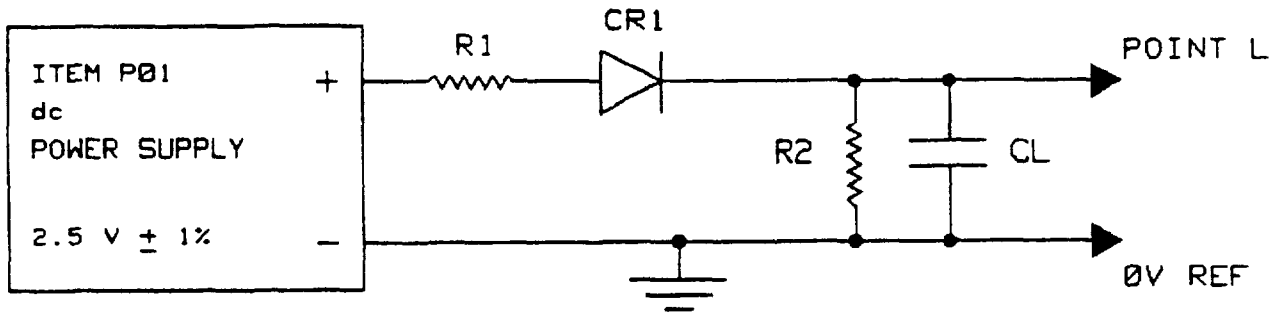


NOTE:

1. Load output under test as specified on figure 13.

FIGURE 12. Output voltage, low and high level, test circuit for tables VIII and IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.



NOTES:

1. CR1 = 1N3600, 1N4150 or equivalent.

2. For all output pins (except pins 52 and 77):
R1 = Select value for a current flow of 12.0 mA ± 5 percent with point L held at 0.4 V.
R2 = 800 Ohm ± 1 percent, 1/8 W.

3. CL = 50 ± 5 pF (including probe and parasitic capacitance) for test conditions Q and R.
CL = 150 ± 15 pF (including probe and parasitic capacitance) for test conditions M1-M64, and N1-N64.
CL = 300 ± 30 pF (including probe and parasitic capacitance) for test conditions M65-M128, and N65-N128.

4. Capacitor CL is not required for dc electrical test conditions G1-G64 and H1-H64.

5. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

6. During output switching, the 2.5 V power supply shall have a maximum undershoot of 500 mV.

7. For pins 52 and 77, R2 = 1256 Ohms ± 1 percent, 1/8 W. The 2.5 V power supply, R1, and CR1 are not required.

FIGURE 13. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE X. Input current, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
J1	D0/CA	21	Open	-800	0	uA	14
J2	D1/CA	70					
J3	D2/CA	22					
J4	D3/CA	71					
J5	D4/CA	23					
J6	D5/CA	72					
J7	D6/CA	24					
J8	D7/CA	73					
J9	D0/CB	28					
J10	D1/CB	78					
J11	D2/CB	29					
J12	D3/CB	79					
J13	D4/CB	30					
J14	D5/CB	80					
J15	D6/CB	81					
J16	D7/CB	82	Open	-800	0	uA	14

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

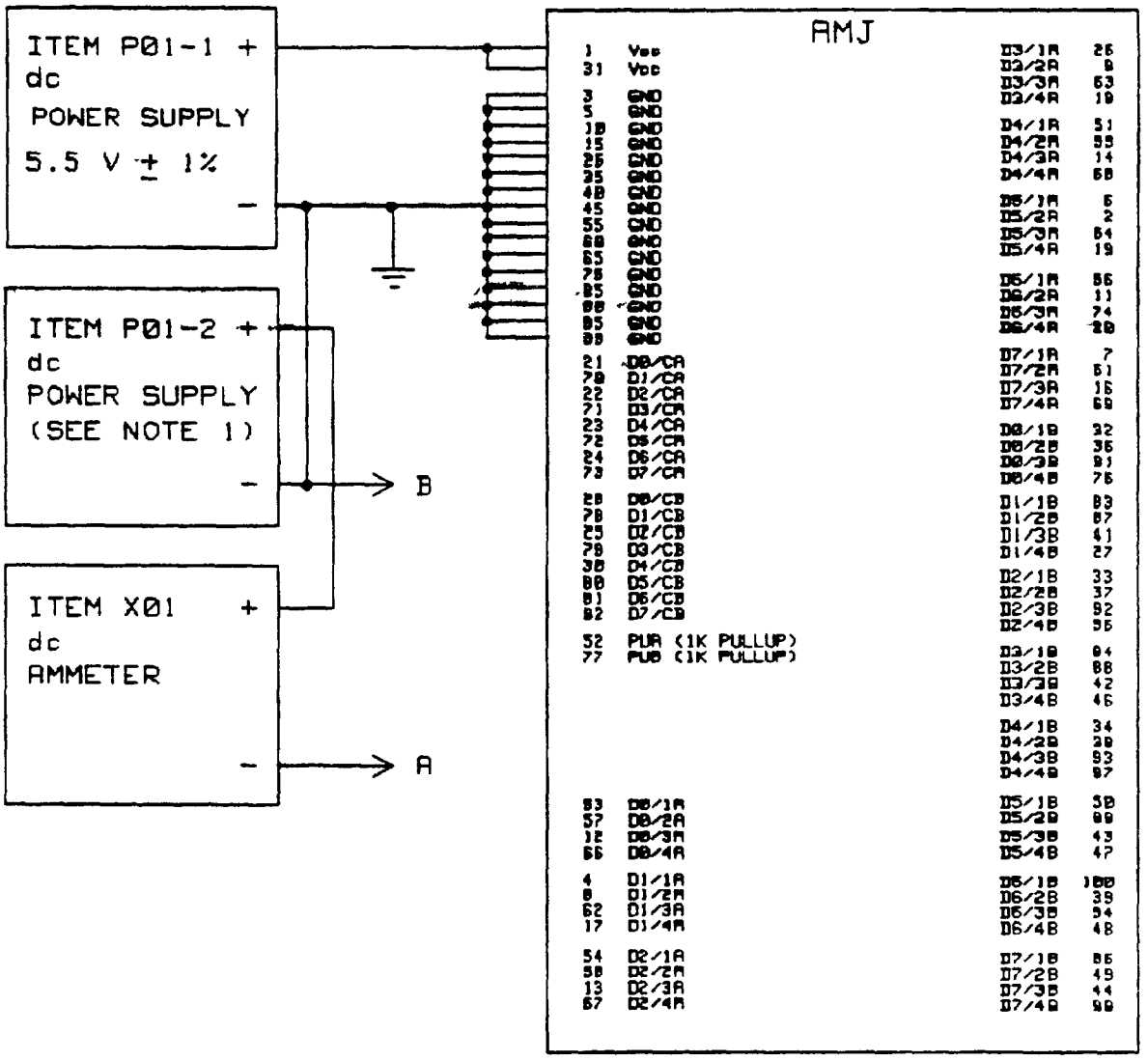
TABLE XI. Input current, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
K1	D0/CA	21	See note 1	0	80	uA	14
K2	D1/CA	70					
K3	D2/CA	22					
K4	D3/CA	71					
K5	D4/CA	23					
K6	D5/CA	72					
K7	D6/CA	24					
K8	D7/CA	73					
K9	D0/CB	28					
K10	D1/CB	78					
K11	D2/CB	29					
K12	D3/CB	79					
K13	D4/CB	30					
K14	D5/CB	80					
K15	D6/CB	81					
K16	D7/CB	82	See note 1	0	80	uA	14

NOTE:

1. One input pin will be connected to point A, while all of the remaining input pins shall be connected to point B. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81 and 82.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.



NOTES:

1. For test conditions:
   J1-J16 - item P01-2 = 0.4 V ± 1 percent
   K1-K16 - item P01-2 = 2.7 V ± 1 percent.

2. Do not isolate the chassis of ammeter X01 from earth ground.

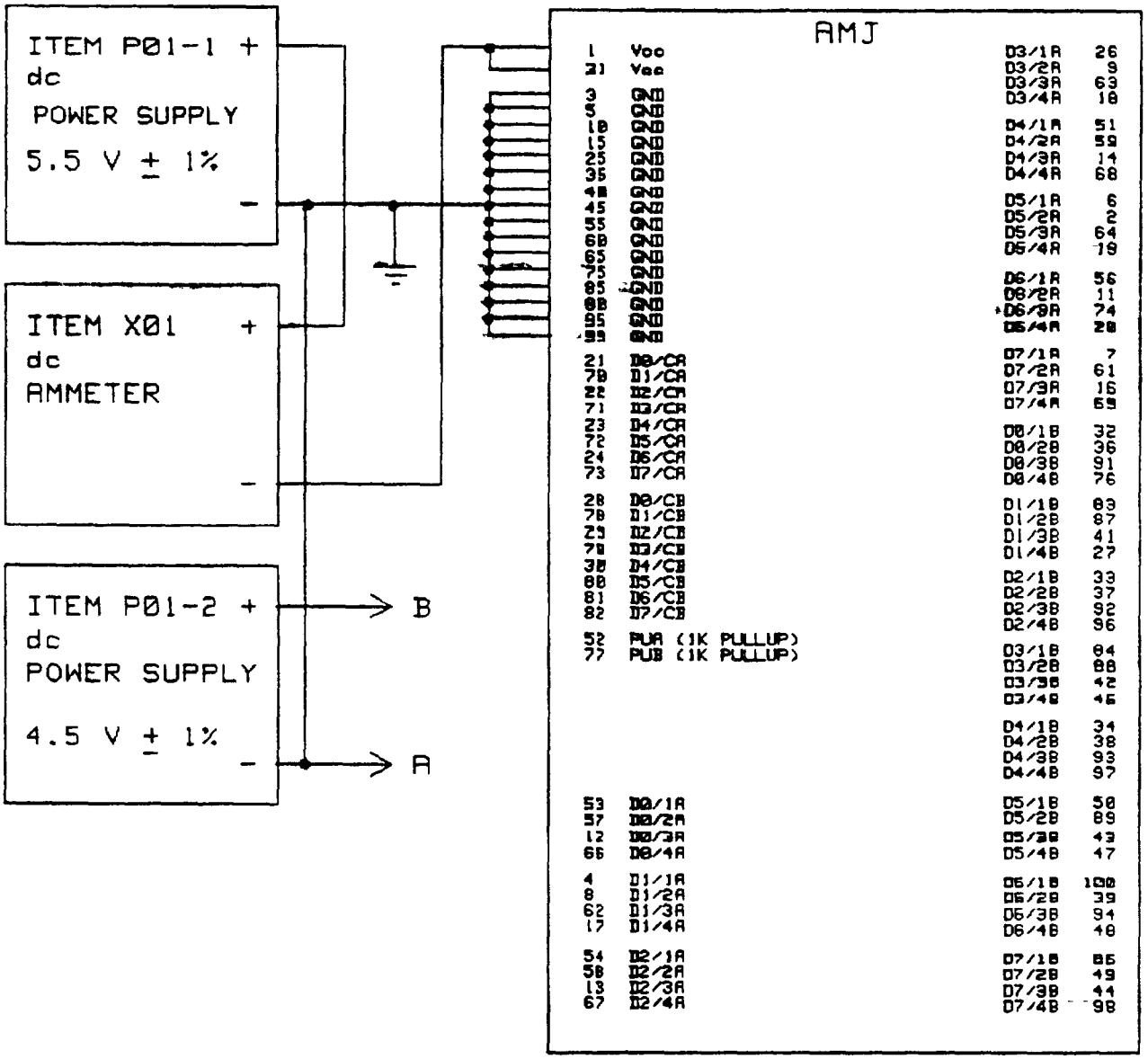
FIGURE 14. Input current, low and high level, test circuit for tables X and XI.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XII. Power supply current test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
L1	Icc (low level)	21,70,22, 71,23,72, 24,73,28, 78,29,79, 30,80,81, 82	Open	60.7	607.0	mA	15
L2	Icc (high level)	Open	21,70,22, 71,23,72, 24,73,28, 78,29,79, 30,80,81, 82	39.1	391.0	mA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.



NOTE:

1. Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 15. Power supply current test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Propagation delay time (tPLH) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
M1	D0/1A	53	21	See note 1	2	24	nS	16,17
M2	D0/2A	57						
M3	D0/3A	12						
M4	D0/4A	66	21					
M5	D1/1A	4	70					
M6	D1/2A	8						
M7	D1/3A	62						
M8	D1/4A	17	70					
M9	D2/1A	54	22					
M10	D2/2A	58						
M11	D2/3A	13						
M12	D2/4A	67	22					
M13	D3/1A	26	71					
M14	D3/2A	9						
M15	D3/3A	63						
M16	D3/4A	18	71					
M17	D4/1A	51	23					
M18	D4/2A	59						
M19	D4/3A	14						
M20	D4/4A	68	23					
M21	D5/1A	6	72	See note 1	2	24	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
M22	D5/2A	2	72	See note 1	2	24	nS	16,17
M23	D5/3A	64						
M24	D5/4A	19	72					
M25	D6/1A	56	24					
M26	D6/2A	11						
M27	D6/3A	74						
M28	D6/4A	20	24					
M29	D7/1A	7	73					
M30	D7/2A	61						
M31	D7/3A	16						
M32	D7/4A	69	73					
M33	D0/1B	32	28					
M34	D0/2B	36						
M35	D0/3B	91						
M36	D0/4B	76	28					
M37	D1/1B	83	78					
M38	D1/2B	87						
M39	D1/3B	41						
M40	D1/4B	27	78					
M41	D2/1B	33	29					
M42	D2/2B	37	29	See note 1	2	24	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
M43	D2/3B	92	29	See note 1	2	24	nS	16,17
M44	D2/4B	96	29					
M45	D3/1B	84	79					
M46	D3/2B	88						
M47	D3/3B	42						
M48	D3/4B	46	79					
M49	D4/1B	34	30					
M50	D4/2B	38						
M51	D4/3B	93						
M52	D4/4B	97	30					
M53	D5/1B	50	80					
M54	D5/2B	89						
M55	D5/3B	43						
M56	D5/4B	47	80					
M57	D6/1B	100	81					
M58	D6/2B	39						
M59	D6/3B	94						
M60	D6/4B	48	81					
M61	D7/1B	86	82					
M62	D7/2B	49						
M63	D7/3B	44	82	See note 1	2	24	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
M64	D7/4B	98	82	See note 1	2	24	nS	16,17
M65	D0/1A	53	21	See note 1	2	33	nS	16,17
M66	D0/2A	57						
M67	D0/3A	12						
M68	D0/4A	66	21					
M69	D1/1A	4	70					
M70	D1/2A	8						
M71	D1/3A	62						
M72	D1/4A	17	70					
M73	D2/1A	54	22					
M74	D2/2A	58						
M75	D2/3A	13						
M76	D2/4A	67	22					
M77	D3/1A	26	71					
M78	D3/2A	9						
M79	D3/3A	63						
M80	D3/4A	18	71					
M81	D4/1A	51	23					
M82	D4/2A	59						
M83	D4/3A	14						
M84	D4/4A	68	23	See note 1	2	33	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the currentversion before use.

TABLE XIII. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
M85	D5/1A	6	72	See note 1	2	33	nS	16,17
M86	D5/2A	2						
M87	D5/3A	64						
M88	D5/4A	19	72					
M89	D6/1A	56	24					
M90	D6/2A	11						
M91	D6/3A	74						
M92	D6/4A	20	24					
M93	D7/1A	7	73					
M94	D7/2A	61						
M95	D7/3A	16						
M96	D7/4A	69	73					
M97	D0/1B	32	28					
M98	D0/2B	36						
M99	D0/3B	91						
M100	D0/4B	76	28					
M101	D1/1B	83	78					
M102	D1/2B	87						
M103	D1/3B	41						
M104	D1/4B	27	78					
M105	D2/1B	33	29	See note 1	2	33	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
M106	D2/2B	37	29	See note 1	2	33	nS	16,17
M107	D2/3B	92	29					
M108	D2/4B	96	29					
M109	D3/1B	84	79					
M110	D3/2B	88						
M111	D3/3B	42						
M112	D3/4B	46	79					
M113	D4/1B	34	30					
M114	D4/2B	38						
M115	D4/3B	93						
M116	D4/4B	97	30					
M117	D5/1B	50	80					
M118	D5/2B	89						
M119	D5/3B	43						
M120	D5/4B	47	80					
M121	D6/1B	100	81					
M122	D6/2B	39						
M123	D6/3B	94						
M124	D6/4B	48	81	See note 1	2	33	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Propagation delay time (tPLH) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
M125	D7/1B	86	82	See note 1	2	33	nS	16,17
M126	D7/2B	49	"	"	"	"	"	"
M127	D7/3B	44	"	"	"	"	"	"
M128	D7/4B	98	82	See note 1	2	33	nS	16,17

NOTE:

1. All of the remaining input pins shall be connected to point C. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81, and 82.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the currentversion before use.

TABLE XIV. Propagation delay time (tPHL) test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N1	D0/1A	53	21	See note 1	2	23	nS	16,17
N2	D0/2A	57						
N3	D0/3A	12						
N4	D0/4A	66	21					
N5	D1/1A	4	70					
N6	D1/2A	8						
N7	D1/3A	62						
N8	D1/4A	17	70					
N9	D2/1A	54	22					
N10	D2/2A	58						
N11	D2/3A	13						
N12	D2/4A	67	22					
N13	D3/1A	26	71					
N14	D3/2A	9						
N15	D3/3A	63						
N16	D3/4A	18	71					
N17	D4/1A	51	23					
N18	D4/2A	59						
N19	D4/3A	14						
N20	D4/4A	68	23					
N21	D5/1A	6	72	See note 1	2	'23	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N22	D5/2A	2	72	See note 1	2	23	nS	16,17
N23	D5/3A	64						
N24	D5/4A	19	72					
N25	D6/1A	56	24					
N26	D6/2A	11						
N27	D6/3A	74						
N28	D6/4A	20	24					
N29	D7/1A	7	73					
N30	D7/2A	61						
N31	D7/3A	16	73					
N32	D7/4A	69	75					
N33	D0/1B	32	28					
N34	D0/2B	36						
N35	D0/3B	91						
N36	D0/4B	76	28					
N37	D1/1B	83	78					
N38	D1/2B	87						
N39	D1/3B	41						
N40	D1/4B	27	78					
N41	D2/1B	33	29					
N42	D2/2B	37	29	See note 1	2	23	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N43	D2/3B	92	29	See note 1	2	23	nS	16,17
N44	D2/4B	96	29					
N45	D3/1B	84	79					
N46	D3/2B	88						
N47	D3/3B	42						
N48	D3/4B	46	79					
N49	D4/1B	34	30					
N50	D4/2B	38						
N51	D4/3B	93						
N52	D4/4B	97	30					
N53	D5/1B	50	80					
N54	D5/2B	89						
N55	D5/3B	43						
N56	D5/4B	47	80					
N57	D6/1B	100	81					
N58	D6/2B	39						
N59	D6/3B	94						
N60	D6/4B	48	81					
N61	D7/1B	86	82					
N62	D7/2B	49	82					
N63	D7/3B	44	82	See note 1	2	23	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N64	D7/4B	98	82	See note 1	2	23	nS	16,17
N65	D0/1A	53	21		2	31	nS	16,17
N66	D0/2A	57						
N67	D0/3A	12						
N68	D0/4A	66	21					
N69	D1/1A	4	70					
N70	D1/2A	8						
N71	D1/3A	62						
N72	D1/4A	17	70					
N73	D2/1A	54	22					
N74	D2/2A	58						
N75	D2/3A	13						
N76	D2/4A	67	22					
N77	D3/1A	26	71					
N78	D3/2A	9						
N79	D3/3A	63						
N80	D3/4A	18	71					
N81	D4/1A	51	23					
N82	D4/2A	59						
N83	D4/3A	14						
N84	D4/4A	68	23	See note 1	2	31	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N85	D5/1A	6	72	See note 1	2	31	nS	16,17
N86	D5/2A	2						
N87	D5/3A	64						
N88	D5/4A	19	72					
N89	D6/1A	56	24					
N90	D6/2A	11						
N91	D6/3A	74						
N92	D6/4A	20	24					
N93	D7/1A	7	73					
N94	D7/2A	61						
N95	D7/3A	16						
N96	D7/4A	69	73					
N97	D0/1B	32	28					
N98	D0/2B	36						
N99	D0/3B	91						
N100	D0/4B	76	28					
N101	D1/1B	83	78					
N102	D1/2B	87						
N103	D1/3B	41						
N104	D1/4B	27	78					
N105	D2/1B	33	29	See note 1	2	31	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N106	D2/2B	37	29	See note 1	2	31	nS	16,17
N107	D2/3B	92						
N108	D2/4B	96	29					
N109	D3/1B	84	79					
N110	D3/2B	88						
N111	D3/3B	42						
N112	D3/4B	46	79					
N113	D4/1B	34	30					
N114	D4/2B	38						
N115	D4/3B	93						
N116	D4/4B	97	30					
N117	D5/1B	50	80					
N118	D5/2B	89						
N119	D5/3B	43						
N120	D5/4B	47	80					
N121	D6/1B	100	81					
N122	D6/2B	39						
N123	D6/3B	94						
N124	D6/4B	48	81					
N125	D7/1B	86	82	See note 1	2	31	nS	16,17

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

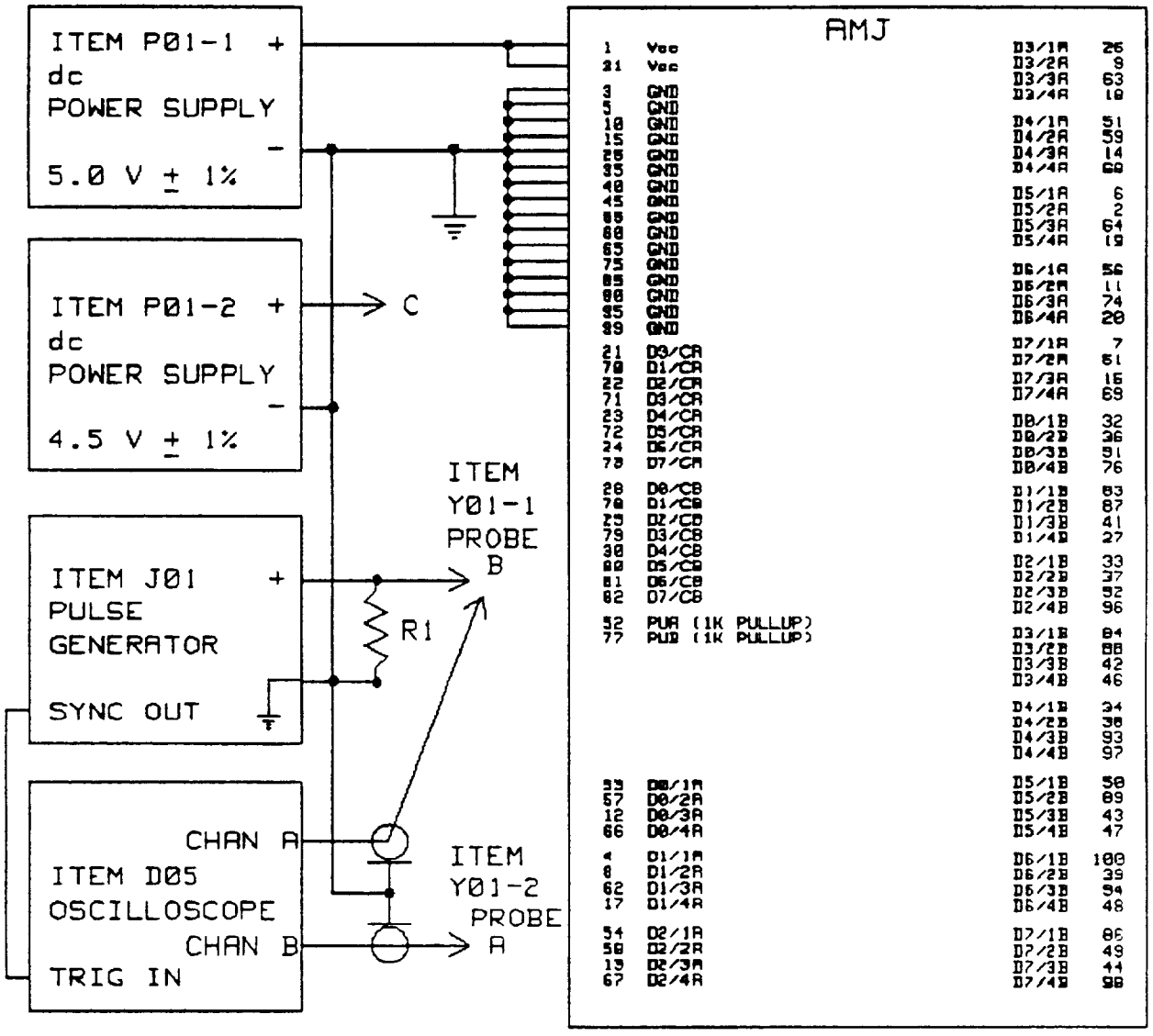
TABLE XIV. Propagation delay time (tPHL) test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
N126	D7/2B	49	82	See note 1	2	31	nS	16,17
N127	D7/3B	44	82	See note 1	2	31	nS	16,17
N128	D7/4B	98	82	See note 1	2	31	nS	16,17

NOTE:

1. All of the remaining input pins shall be connected to point C. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81, and 82.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

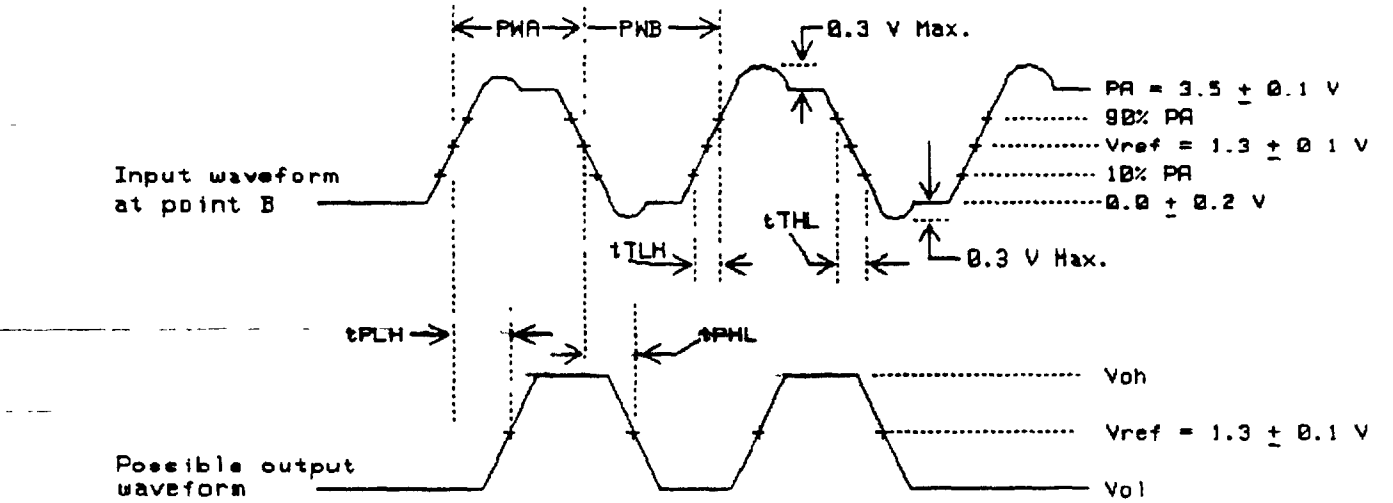


NOTES:

1. R1 = 51 Ohms ± 5 percent, 1 W.
2. Load each output as specified on figure 13.

FIGURE 16. Propagation delay (tPLH and tPHL) test circuit for tables XIII and XIV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters: PWA = PWB = 500 nS ± 10 percent, tTLH ≤ 15 nS, tTHL ≤ 6 nS.

2. Before taking the measurement, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V.

FIGURE 17. Propagation delay time (tPLH and tPHL) test circuit waveforms for tables XIII and XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XV. Output crosstalk, low level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
P1	D0/1A	53	Open	21	See note 1	0	400	mV	18, 19
P2	D0/2A	57							
P3	D0/3A	12							
P4	D0/4A	66		21					
P5	D1/1A	4		70					
P6	D1/2A	8							
P7	D1/3A	62							
P8	D1/4A	17		70					
P9	D2/1A	54		22					
P10	D2/2A	58							
P11	D2/3A	13							
P12	D2/4A	67		22					
P13	D3/1A	26		71					
P14	D3/2A	9							
P15	D3/3A	63	Open	71	See note 1	0	400	mV	18, 19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XV. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
P16	D3/4A	18	Open	71	See note 1	0	400	mV	18,19
P17	D4/1A	51		23					
P18	D4/2A	59							
P19	D4/3A	14							
P20	D4/4A	68		23					
P21	D5/1A	6		72					
P22	D5/2A	2							
P23	D5/3A	64							
P24	D5/4A	19		72					
P25	D6/1A	56		24					
P26	D6/2A	11							
P27	D6/3A	74							
P28	D6/4A	20		24					
P29	D7/1A	7		73					
P30	D7/2A	61	Open	73	See note 1	0	400	mV	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XV. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
P31	D7/3A	16	Open	73	See note 1	0	400	mV	18,19
P32	D7/4A	69		73					
P33	D0/1B	32		28					
P34	D0/2B	36							
P35	D0/3B	91							
P36	D0/4B	76		28					
P37	D1/1B	83		78					
P38	D1/2B	87							
P39	D1/3B	41							
P40	D1/4B	27		78					
P41	D2/1B	33		29					
P42	D2/2B	37							
P43	D2/3B	92							
P44	D2/4B	96		29					
P45	D3/1B	84	Open	79	See note 1	0	400	mV	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XV. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
P46	D3/2B	88	Open	79	See note 1	0	400	mV	18,19
P47	D3/3B	42		79					
P48	D3/4B	46		79					
P49	D4/1B	34		80					
P50	D4/2B	38							
P51	D4/3B	93							
P52	D4/4B	97		80					
P53	D5/1B	50		80					
P54	D5/2B	89							
P55	D5/3B	43							
P56	D5/4B	47		80					
P57	D6/1B	100		81					
P58	D6/2B	39							
P59	D6/3B	94							
P60	D6/4B	48	Open	81	See note 1	0	400	mV	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XV. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
P61	D7/1B	86	Open	82	See note 1	0	400	mV	18, 19
P62	D7/2B	49							
P63	D7/3B	44							
P64	D7/4B	98	Open	82	See note 1	0	400	mV	18, 19

NOTE:

1. All of the remaining input pins shall be connected to point C. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81 and 82.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk, high level, test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
Q1	D0/1A	53	21	Open	See note 1	0	400	mV	18,19
Q2	D0/2A	57							
Q3	D0/3A	12							
Q4	D0/4A	66	21						
Q5	D1/1A	4	70						
Q6	D1/2A	8							
Q7	D1/3A	62							
Q8	D1/4A	17	70						
Q9	D2/1A	54	22						
Q10	D2/2A	58							
Q11	D2/3A	13							
Q12	D2/4A	67	22						
Q13	D3/1A	26	71						
Q14	D3/2A	9							
Q15	D3/3A	63	71	Open	See note 1	0	400	mV	18,19

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Q16	D3/4A	18	71	Open	See note 1	0	400	mV	18, 19
Q17	D4/1A	51	23						
Q18	D4/2A	59							
Q19	D4/3A	14							
Q20	D4/4A	68	23						
Q21	D5/1A	6	72						
Q22	D5/2A	2							
Q23	D5/3A	64							
Q24	D5/4A	19	72						
Q25	D6/1A	56	24						
Q26	D6/2A	11							
Q27	D6/3A	74							
Q28	D6/4A	20	24						
Q29	D7/1A	7	73						
Q30	D7/2A	61	73	Open	See note 1	0	400	mV	18, 19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
Q31	D7/3A	16	73	Open	See note 1	0	400	mV	18,19
Q32	D7/4A	69	73						
Q33	D0/1B	32	28						
Q34	D0/2B	36							
Q35	D0/3B	91							
Q36	D0/4B	76	28						
Q37	D1/1B	83	78						
Q38	D1/2B	87							
Q39	D1/3B	41							
Q40	D1/4B	27	78						
Q41	D2/1B	33	29						
Q42	D2/2B	37							
Q43	D2/3B	92							
Q44	D2/4B	96	29						
Q45	D3/1B	84	79	Open	See note 1	0	400	mV	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Q46	D3/2B	88	79	Open	See note 1	0	400	mV	18, 19
Q47	D3/3B	42							
Q48	D3/4B	46	79						
Q49	D4/1B	34	30						
Q50	D4/2B	38							
Q51	D4/3B	93							
Q52	D4/4B	97	30						
Q53	D5/1B	50	80						
Q54	D5/2B	89							
Q55	D5/3B	43							
Q56	D5/4B	47	80						
Q57	D6/1B	100	81						
Q58	D6/2B	39							
Q59	D6/3B	94							
Q60	D6/4B	48	81	Open	See note 1	0	400	mV	18, 19

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T18:11Z
Check the source to verify that this is the current version before use.

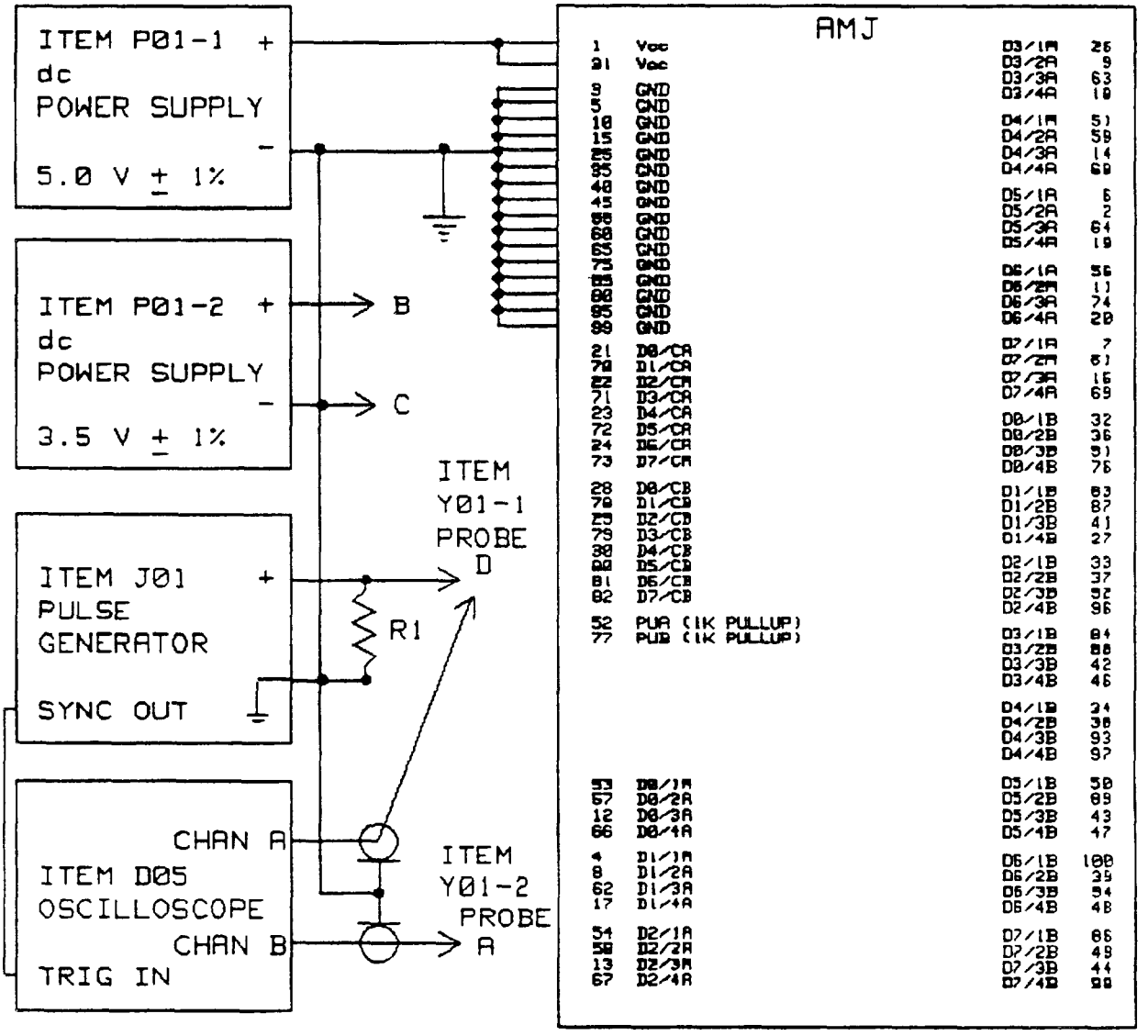
TABLE XVI. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Q61	D7/1B	86	82	Open	See note 1	0	400	mV	18, 19
Q62	D7/2B	49							
Q63	D7/3B	44							
Q64	D7/4B	98	82	Open	See note 1	0	400	mV	18, 19

NOTE:

1. One input pin will be connected to point B while all of the remaining input pins shall be connected to point D. Input pins are 21, 70, 22, 71, 23, 72, 24, 73, 28, 78, 29, 79, 30, 80, 81 and 82.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:11Z
Check the source to verify that this is the current version before use.

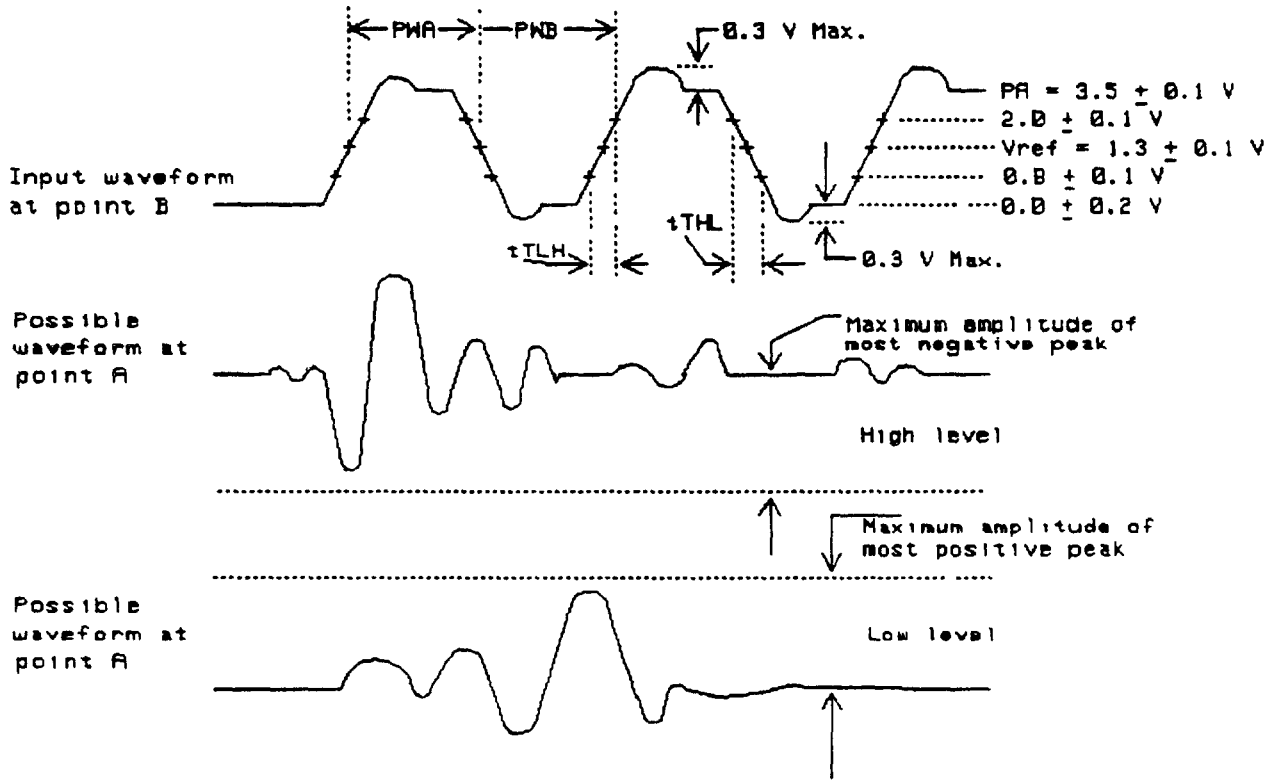


NOTES:

1. R1 = 51 Ohms ± 5 percent, 1 W.
2. Load each output as specified on figure 13.

FIGURE 18. Output crosstalk, low and high level, test circuit for tables XV and XVI.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the currentversion before use.



NOTE:

1. Input pulse parameters: PWA = PWB = 250 nS ± 10 percent, tTLH = tTHL = 3.5 ± 1.5 nS (verified at each input).

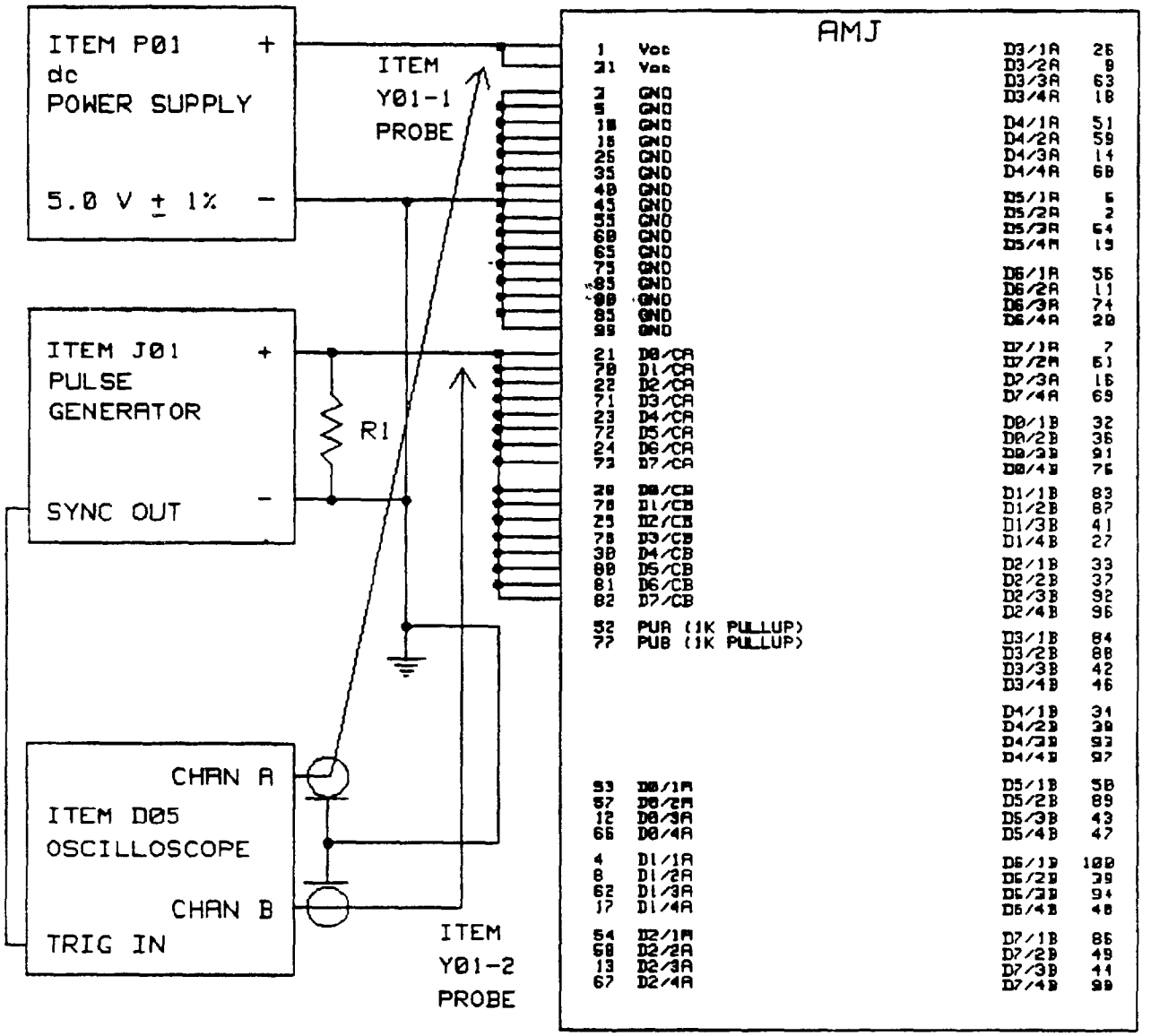
FIGURE 19. Output crosstalk, low and high level, test circuit for tables XV and XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XVII. Power supply transient amplitude test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Figs
		Min	Max		
R	PSTA	0	400	mV	20,21

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

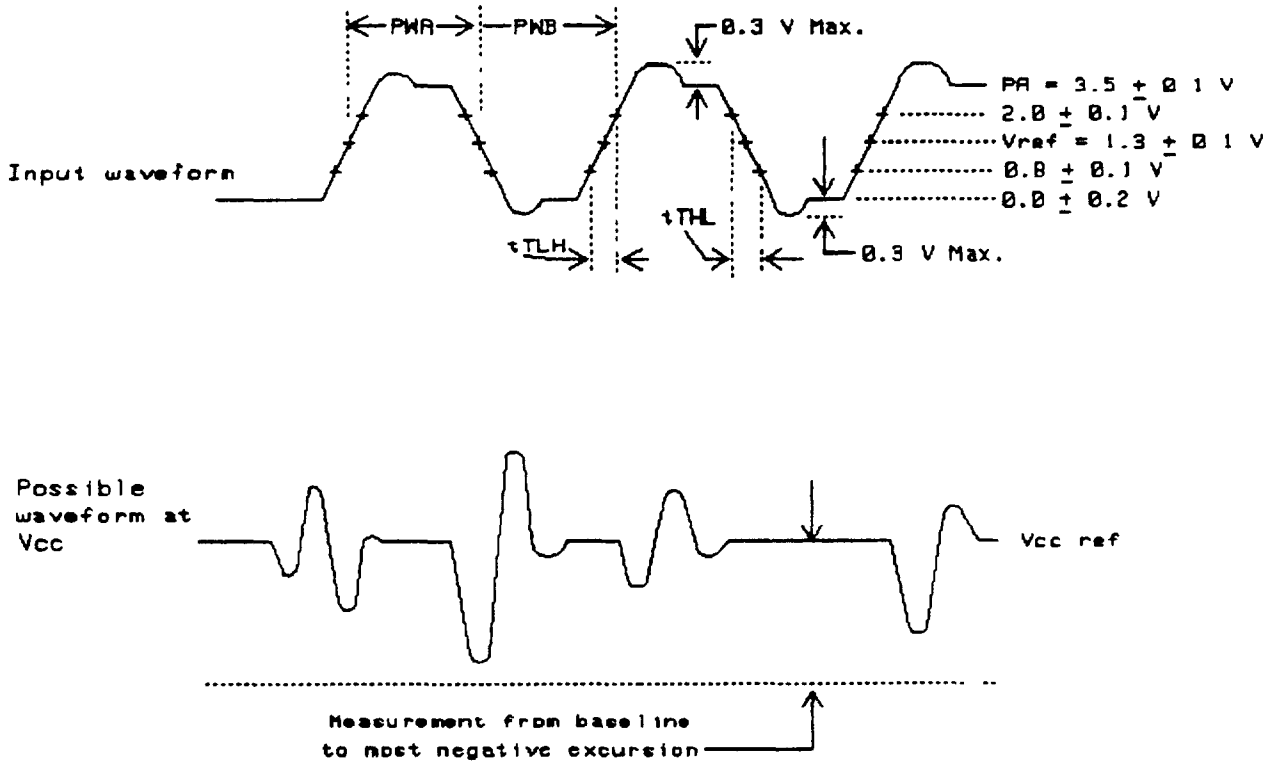


NOTE:

1. R1 = 51 Ohms ± 5 percent, 1 W.

FIGURE 20. Power supply transient amplitude test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters PWA = PWB = 250 nS ± 10 percent, tTLH ≤ 15 nS, tTHL ≤ 6 nS.

2. Connect point A as close as possible to pin 1 on the module and observe power supply transient on item D05. Typical waveforms are shown above.

FIGURE 21. Power supply transient amplitude waveform for table XVII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply filter capacitance test.

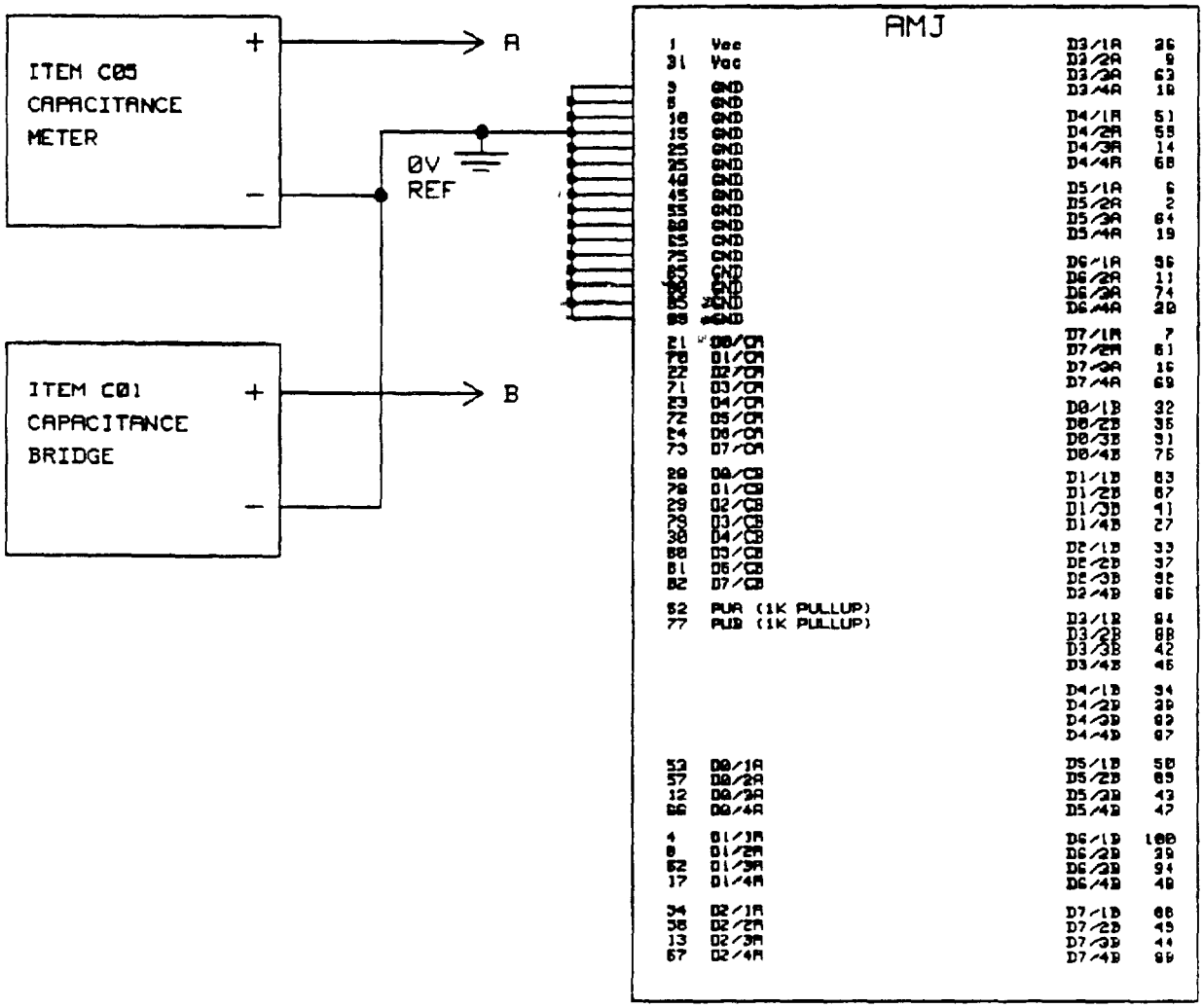
Test cond	Parameter	B to pin	25 ± 5°C Initial limits Min	Max	Units	Fig
S1	PSFC	1,31	3.85	7.15	uF	22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.

TABLE XIX. Input capacitance test.

Test cond	Signal name	Connect A to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
T1	D0/CA	21	2	55	pF	22
T2	D1/CA	70				
T3	D2/CA	22				
T4	D3/CA	71				
T5	D4/CA	23				
T6	D5/CA	72				
T7	D6/CA	24				
T8	D7/CA	73				
T9	D0/CB	28				
T10	D1/CB	78				
T11	D2/CB	29				
T12	D3/CB	79				
T13	D4/CB	30				
T14	D5/CB	80				
T15	D6/CB	81				
T16	D7/CB	82	2	55	pF	22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:11Z
Check the source to verify that this is the current version before use.



NOTES:

1. Null out the effect of the test leads prior to connecting to the module.

2. Do not attach test connector or wires to any of the module pins other than the pin under test. Stray capacitance added to other pins may affect the readings.

3. Capacitance bridge (C01) setup:
   Frequency = 1 KHz.
   Vpk = 0.3 V maximum.

FIGURE 22. Power supply filter and input capacitance test circuit for tables XVIII and XIX.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7.

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-M-28787/393	2 DOCUMENT DATE (YYMMDD) 24 August 1990

3 DOCUMENT TITLE
Modules, Digital, Standard Electronic Interface, Bus Module, Key Code AMJ

4 NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.*)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a. NAME . James Reeves (Code 06QM2)	b. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (703) 746-1926
c. ADDRESS (*Include Zip Code*) Commander, Naval Sea Systems Department of the Navy Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

Source: https://assist.dla.mil - Downloaded: 2026-04-20T18:11Z
Check the source to verify that this is the current version before use.